

Keeping Rare Classes Reachable: Class-Conditional Label Narrowing for In-Context Classification

Anonymous ACL submission

Abstract

In-context classification with small large language models (LLMs) is hard when there are many classes and the labelled data are skewed. A small classifier can narrow the label set before prompting, which raises the question of how it should narrow. We compare narrowing rules built from the same classifier on six English datasets with eight open LLMs, pairing every comparison instance by instance. When the labelled pool is long-tailed, class-conditional conformal sets, as used by Conformal In-Context Learning (CICLe), keep about 90% coverage or more on every class, while top- k , probability-mass and marginal conformal sets lose the rarest classes and 4 to 13 macro-F1 points. Narrowing also gives a small, reliable gain over few-shot prompting, 0.5 to 1.8 points with 11 to 45% shorter prompts, largest when label names carry no meaning. With plentiful balanced labels a fine-tuned encoder is the better choice. Comparisons across prompting methods must also hold the label information constant, since prompts that show only a few labels leave up to 65% of outputs unparseable.

1 Introduction

Small open large language models (LLMs) can classify text from a few labelled examples placed in the prompt (Brown et al., 2020), which makes them attractive when labelled data are scarce. Their accuracy depends on which examples are shown and in what order (Liu et al., 2022; Lu et al., 2022; Zhao et al., 2021), and real labelled data make it harder. With many classes the prompt must carry a long label list, and with a skewed pool the rare classes have few examples to show.

A natural remedy is to let a smaller model, such as a classifier trained on the labelled pool, narrow the label space before prompting, so that the LLM chooses among a few candidates with examples from those classes only (Milios et al., 2023; Randl

et al., 2024). This paper asks how that classifier should narrow. Keeping its top- k labels is the obvious choice, but a classifier trained on a skewed pool is least reliable on the rare classes, and a label left out of the set cannot be predicted. Conformal In-Context Learning (CICLe) (Randl et al., 2024) narrows instead with class-conditional conformal prediction, which calibrates one threshold per class so that every class, frequent or rare, stays in the set with a chosen probability. Our main finding is that this calibration is what keeps rare classes reachable. Under a long-tailed labelled pool, the worst-covered class keeps 89% coverage under class-conditional sets, while top- k , probability-mass and marginal conformal sets of the same size cover the rarest class for at most 36% of its instances, and often for none.

CICLe was evaluated on one food-safety dataset with Per-Class retrieval, which shows every candidate class among the examples, so its prompt need not list the labels. Fixed retrieval and zero-shot and few-shot baselines show few or no labels, so comparing them requires the label list in every prompt. We evaluate it and the alternative rules on six English datasets with six open LLMs of 3B to 8B parameters and two of 12B and 32B. Our contributions are the following, main finding first.

1. **How you narrow matters.** On balanced data, class-conditional conformal sets match top- k , probability-mass and marginal sets of the same size. Under a long-tailed labelled pool only class-conditional calibration keeps the rare classes in the candidate set, and the alternatives lose 4 to 13 points that the LLM cannot recover (Section 5.3).
2. **Narrowing gives a small gain and a shorter prompt.** CICLe beats few-shot prompting in nine of ten dataset and retrieval cells by 0.5 to 1.8 macro-F1 points and shortens Per-Class prompts by 11 to 45%, and its gain reaches 6.5

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points when label names are nonsense words (Sections 5.2 and 5.4).

3. **When the LLM pipeline is worth it.** With 1,600 balanced labels a fine-tuned encoder or the base classifier beats every 3B to 8B pipeline. Under a long-tailed or small pool the order flips, and the miscoverage level should follow the strength of the classifier (Sections 5.5 and 5.6).
4. **Comparisons must hold the label information constant.** A protocol for comparing prompting methods with the admissible labels in every prompt, a shared labelled pool, seeded conformal sets and instance-level paired tests. When prompts show only the retrieved labels, as with Fixed retrieval, up to 65% of a model’s outputs do not parse, and the parse failures decide which models appear to benefit (Section 5.1).

After the background and setup, Section 5 first establishes the measurement (Sections 5.1 and 5.2) and then turns to the findings above. Code, prompts and per-instance records will be released.

2 Related Work

Few-shot prompting and example selection. In-context learning lets an LLM classify from labelled examples in the prompt (Brown et al., 2020). Its accuracy depends on which examples are shown and in what order (Lu et al., 2022), and on biases toward frequent or recent labels that can be corrected after the fact (Zhao et al., 2021). Retrieving similar examples is the standard remedy (Liu et al., 2022; Rubin et al., 2022), and selection can be cast as a search problem (Purohit et al., 2025). We keep similarity-based retrieval fixed and vary only which classes may contribute examples.

Two-stage pipelines. Cascades send an input to a stronger model only when a weaker one is not confident (Varshney and Baral, 2022; Chen et al., 2024; Yue et al., 2024), and routers learn that decision from data (Ding et al., 2024). Fisch et al. (2021) conformalise such a cascade so that implausible labels are pruned early with a coverage guarantee. For classification with many labels, a retriever can show the LLM only part of the label space (Milios et al., 2023), or the LLM can first generate candidate labels that are then reranked (Zhu and Zamani, 2024; D’Oosterlinck et al., 2024). CICLE (Randl et al., 2024) belongs to this family, with a

calibrated classifier as the first stage, and our comparison of narrowing rules asks which part of the design carries the benefit.

Conformal prediction in NLP and for LLMs. Conformal prediction turns any scoring model into a set-valued predictor with a finite-sample coverage guarantee (Vovk et al., 2005; Angelopoulos and Bates, 2023). Its split form needs one held-out calibration set (Papadopoulos et al., 2002), and calibrating per class gives coverage within each class rather than on average (Vovk, 2012). In NLP it has been applied to classifiers on pretrained encoders (Maltoudoglou et al., 2020) and to LLM outputs, for multiple-choice answers (Kumar et al., 2023), generated text (Quach et al., 2024), models without logit access (Su et al., 2024) and deciding when an agent should ask for help (Ren et al., 2023), as surveyed by Campos et al. (2024). These methods calibrate the LLM itself. CICLE calibrates a small classifier and uses its sets to shape the prompt.

Labels in the prompt, and prompting versus fine-tuning. Min et al. (2022) found that random labels barely hurt and that showing the label space mattered more, while Yoo et al. (2022) found the effect of correct labels to depend on model and task. With flipped or semantically unrelated labels, the ability to learn the new mapping grows with model size, and small models fall back on their priors about the label words (Wei et al., 2023; Pan et al., 2023; Kossen et al., 2024). Our label renaming uses this setting as a stress test with 3B to 8B models. Several evaluations report that a fine-tuned encoder still beats few-shot LLM prompting when enough labelled data exist (Edwards and Camacho-Collados, 2024; Bucher and Martini, 2024; Yu et al., 2023; Sun et al., 2023). We include such baselines and ask under which data conditions the LLM pipeline pays off.

3 Background: Narrowing the Label Space with Conformal Prediction

Split conformal prediction, in plain terms. Take any classifier that scores each class. Hold out a calibration split it has not seen and record, for each calibration example, how low the classifier scored the true label. For a new input, include in the candidate set every class whose score is no worse than what the true label received on all but a fraction α of the calibration examples. If the new input is exchangeable with the calibration split, the

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true label is in the set with probability at least $1 - \alpha$, for any classifier and any sample size (Vovk et al., 2005; Papadopoulos et al., 2002). A weak classifier does not lower the coverage, it makes the sets larger.

Class-conditional calibration. The guarantee above is marginal. On a long-tailed dataset the set can cover 95% of inputs overall while missing most inputs of a rare class, because the rare class contributes few calibration examples to the shared threshold. Class-conditional conformal prediction computes one threshold per class from the calibration examples of that class only (Vovk, 2012), so coverage holds within each class. Top- k selection and probability thresholds give no such guarantee, the first because it ignores the classifier’s confidence, the second because it trusts a confidence that is rarely calibrated.

The CICLE pipeline. CICLE (Randl et al., 2024) trains a base classifier (logistic regression on sentence embeddings) on the labelled pool and calibrates it class-conditionally. For a test input it yields a candidate set $\mathcal{C}(x)$. If $|\mathcal{C}(x)| = 1$ that label is returned without calling the LLM, and if the set is empty the classifier’s top label is. Otherwise the LLM is prompted with the task description, the candidate labels and labelled examples retrieved from the candidate classes only.

Narrowing rules compared. To separate narrowing as such from conformal calibration, we build candidate sets from the same classifier with five rules. *CP* is the class-conditional conformal set at miscoverage α . *Marginal CP* shares one threshold across classes. *Top- m* keeps the m most probable classes, where m is the mean CP set size on that dataset and seed, so the budget matches but the size does not adapt to the input. *Probability mass* keeps the most probable classes until their predicted probabilities sum to $1 - \alpha$, which adapts but trusts the uncalibrated confidence. *Oracle* contains the gold label plus $m - 1$ random classes and bounds any rule at that size. All rules share the prompt, the retrieval and the singleton and empty-set behaviour.

Fixed and Per-Class retrieval. *Fixed* retrieval places the k training examples most similar to the input (cosine similarity in the embedding space) among those whose label is in the candidate set. *Per-Class* retrieval places the k most similar examples of each candidate class, so the prompt holds

Dataset	Domain	Cls.	Imb.	Len.	Variants
SST-5	Sentiment	5	$2.1\times$	23	I, P, R
Yahoo Answers	Topic	10	$1.4\times$	50	I, P, R
SemEval-18	Emoji	20	$8.7\times$	20	R
GoEmotions	Emotion	28	$329\times$	16	R
Ohsumed	Medical	23	$65\times$	271	R, F
MASSIVE	Intent	59	$58\times$	8	R

Table 1: Datasets. *Cls.* is the number of classes, *Imb.* the ratio between the largest and the smallest class in the full training set, *Len.* the mean input length in Llama-3.1 tokens. *Variants*: I = controlled imbalance of the labelled pool ($10\times$, $100\times$), P = pool size (250, 500, 1,000), R = label renaming, F = Fixed retrieval only beyond $k = 1$ (a Per-Class prompt at $k = 4$ would hold about 90 abstracts). Every dataset is subsampled to 1,600 training, 400 calibration and 1,000 test instances per seed (Section 4.3).

$k \times |\mathcal{C}(x)|$ examples and shrinks with the set. Few-shot prompting is the case in which the candidate set is the full label set. Narrowing also makes listing labels affordable, since few-shot prompting must list every class and CICLE only the candidate set.

4 Experimental Setup

4.1 Datasets

Table 1 lists the six English single-label datasets, chosen to vary the number of classes, the class imbalance, the input length and the information carried by the label names. SST-5 (Socher et al., 2013) is the near-balanced five-class case, Yahoo Answers (Zhang et al., 2015) has ten balanced topics with informative names and a strong base classifier, SemEval-18 Task 2 (Barbieri et al., 2018) predicts the emoji of a tweet, so its twenty label names carry little meaning, GoEmotions (Demszky et al., 2020), restricted to single-label comments, is the many-class long-tailed case, and Ohsumed (Hersh et al., 1994) classifies medical abstracts of about 270 tokens into 23 disease categories. MASSIVE (FitzGerald et al., 2023) intent classification, with 59 intents of voice-assistant commands of about seven words, adds a second naturally long-tailed dataset with many classes and short inputs from a different domain (Appendix J). The two balanced datasets serve as the substrate for the controlled-imbalance and pool-size experiments, since the skewed ones cannot be rebalanced without changing the task.

4.2 Models

The main grid uses six open instruction-tuned LLMs, two sizes from each of three families: Llama-3.2-3B and Llama-3.1-8B (Grattafiori et al., 2024), Qwen2.5-3B and Qwen2.5-7B (Yang et al., 2024), Ministral-3-3B (Mistral AI, 2025) and Mistral-7B-v0.3 (Jiang et al., 2023). Mistral-NeMo-12B (Mistral AI and NVIDIA, 2024) and Qwen2.5-32B (Yang et al., 2024) are run with and without CICLE on the five short-text datasets (Appendix H). All run in bfloat16 with greedy decoding through their chat templates. The base classifier of every narrowing method is logistic regression on all-MiniLM-L6-v2 sentence embeddings (Reimers and Gurevych, 2019; Wang et al., 2020), and the same embeddings retrieve the examples by cosine similarity, so MiniLM is both the classifier’s feature space and the retriever. Calibration uses crepes (Boström, 2022) with seeded tie-breaking. Contriever (Izacard et al., 2022), TF-IDF and an SVM appear in the ablations.

4.3 Protocol

Every experiment uses a stratified subsample of 2,000 labelled and 1,000 test instances per dataset, drawn with three seeds (42, 43, 44). Of the 2,000, 400 form the calibration split, the proportion of Randl et al. (2024), and the base classifier, the retrieval pool and the supervised baselines all see the same 1,600 training examples. One test sample per seed is shared by every method, which makes paired tests over instances possible, and the seeds change the subsample, the calibration split and the conformal tie-breaking (Appendix A gives the reasoning behind the sizes).

The released CICLE implementation lists no labels, which suits its Per-Class retrieval. Because we add Fixed retrieval and zero-shot and few-shot baselines, the protocol holds the label information constant and changes five things. Every prompt lists the admissible labels, all classes or the candidate set, with identical wording otherwise. Every method retrieves from the same pool. Calibration split, conformal sets and test sample are seeded. Outputs are normalised before matching, and anything that still matches no label counts as wrong and is reported as an invalid output. A per-instance record is saved for every run. Appendix A gives the template, the parser and the original prompt format, which we rerun inside the same framework (Appendix B).

Both retrieval variants are run for every prompting method, with $k \in \{1, 2, 4, 8\}$ on the five short-text datasets and $k \in \{1, 4\}$ elsewhere, and Per-Class retrieval on Ohsumed at $k = 1$ only (Appendix A). The reference setting is MiniLM, logistic regression and $\alpha = 0.05$, the choices of the original paper.

4.4 Experiments

Each experiment beyond the main grid isolates one variable. *Narrowing rules*: the four alternative rules of Section 3 on every dataset and variant, $k \in \{1, 4\}$. *Controlled imbalance*: on SST-5 and Yahoo Answers the training sample is replaced by a long-tailed one whose class sizes decay geometrically to a ratio of 10 or 100, with at least five examples per class and a class order drawn from the seed, while the test sample is unchanged (Appendix J). *Label renaming*: on all six datasets every class name is replaced by a nonsense word, the same for all seeds, leaving texts, splits and classifier unchanged. *No examples*: CICLE with $k = 0$ shows the candidate list alone. *Pool size*: on SST-5 and Yahoo Answers the labelled pool is reduced to 250, 500 and 1,000 examples, split 80/20 as before, with every pipeline and baseline rerun. *Ablations*: $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$, the embedding and the classifier, with Llama-3.1-8B and Llama-3.2-3B on the short-text datasets. *Prompt robustness*: a second template with the labels in random order and the most similar example last, and random instead of similarity-based retrieval, on seed 42 with $k \in \{1, 4\}$.

Roadmap. The main claim rests on the alternative rules under controlled and natural imbalance (Section 5.3), the gain claim on the main grid, $k = 0$ and label renaming (Sections 5.2 and 5.4), the claim about when the LLM is worth it on the baselines, pool sizes, α and larger models (Sections 5.5 and 5.6), and the label-information claim on the rerun of the original prompt format, which Section 5.1 reports first because every other comparison depends on it.

4.5 Methods and Baselines

Zero-shot prompting lists all labels and no examples, *few-shot* prompting adds retrieved examples from all classes, and CICLE restricts both to the class-conditional conformal set, returning a singleton set without an LLM call and the classifier’s top label for an empty set. The alternative narrowing

rules differ from CICLE only in the set. Three supervised baselines use no LLM, namely the base classifier itself, and RoBERTa-base and RoBERTa-large (Liu et al., 2019) fine-tuned on the same 1,600 examples for ten epochs (AdamW, learning rate 2×10^{-5} for base and 10^{-5} for large, batch size 16, 256 tokens, linear warm-up and decay) with the epoch chosen on the 400 calibration examples. The recipe is not tuned further, so the encoder scores are a floor. All baselines are run for three seeds and every dataset variant.

4.6 Metrics and Statistics

We report macro-F1 over the classes present in the test sample (accuracy is in Appendix K), the invalid-output rate, the mean prompt tokens per LLM call, the share of instances the LLM never sees, and the coverage, mean size and lowest per-class coverage of the candidate sets. Every comparison between two methods is paired, with both sharing the model, seed, k and retrieval variant, hence the same test instances, pool and conformal sets. We report the mean paired macro-F1 difference over all such cells with a 95% interval from a bootstrap with 5,000 resamples of the test instances (Koehn, 2004; Dror et al., 2018), with the same resample applied to both methods in every cell, so the interval reflects test-sample noise and not the spread over hyperparameters. The number of pairs is 72 for the main grid (six models, three seeds, four values of k), 36 where $k \in \{1, 4\}$, 12 or 6 per model and 12 for the seed-42 checks. A difference is significant when its interval excludes zero, printed in bold in the tables, otherwise marked n.s.

5 Results

All numbers are at the reference setting and averaged over the six small models and three seeds unless stated. A gain is CICLE minus an alternative in macro-F1 points, with a 95% paired bootstrap interval where it is the key number.

5.1 Comparisons Must Hold the Label Information Constant

A comparison of prompting methods must hold the label information constant, because prompts that show few labels produce unparseable outputs that decide the comparison. The prompt format of the released CICLE implementation lists no labels, which suits its Per-Class retrieval, and we ran it with Fixed retrieval, which shows only the retrieved labels, on seed 42 (Table 4, Appendix B). Up to

65% of a model’s few-shot outputs then matched no label, and the per-model effect of narrowing changed sign between the two formats in a third of the cells. Under the corrected protocol Per-Class CICLE never significantly hurts any of the six models (Table 5, Appendix C), which leaves the question of how large its gain is.

5.2 Narrowing Gives a Small Gain and a Shorter Prompt

Narrowing the label set gives a small, reliable gain over few-shot prompting and a shorter prompt. Table 2 compares zero-shot, few-shot and CICLE prompting at $k = 4$ on the six datasets, with the paired gain pooled over all k in panel B, and Figure 1 plots every configuration against its prompt length. CICLE is ahead in nine of ten dataset and retrieval cells, with gains between 0.5 and 1.8 points whose intervals exclude zero (Table 2B). The exception is Fixed retrieval on Yahoo Answers, where one retrieved example already carries the label and the difference is not significant. The gains do not grow with k (Appendix K). The one large gain is Ohsumed with Per-Class retrieval, +5.2 [+4.4, +6.1] points, where a few-shot prompt holds one long abstract for every class and the LLM loses the input among them, while CICLE halves the list. The candidate list alone, with no examples, is worth about one retrieved example (Table 2A, row $k = 0$).

The saving is on the prompt. Per-Class retrieval places k examples per admissible class, so a narrower list shortens the prompt in proportion, by 11 to 45% at equal k (Table 2B), at the same or higher macro-F1, which is why the filled blue markers of Figure 1 lie above and to the left of the grey ones. Fixed retrieval is cheaper still, and on GoEmotions it also scores higher, because examples for most of its 28 classes dilute the prompt more than they inform it. If narrowing helps, the next question is whether the rule used to narrow matters.

5.3 How You Narrow Matters

How one narrows matters only when the labelled pool is long-tailed. Figure 2 compares CICLE with top- m , probability-mass and marginal conformal sets built from the same classifier at the same mean size on Yahoo Answers and SST-5, whose labelled pool is resampled to a 10 or 100 times imbalance while the test sample stays fixed, and Table 6 (Appendix D) gives every comparison. On balanced data the four rules are interchangeable, and no dif-

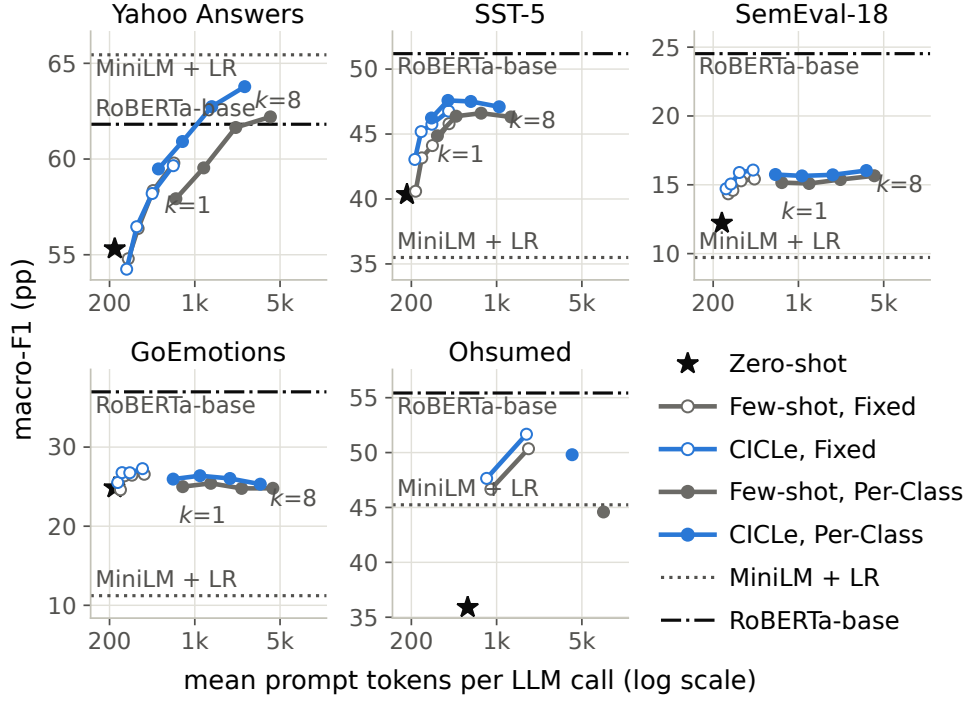


Figure 1: Macro-F1 against mean prompt tokens per LLM call (log scale), one panel per dataset. Each point is one method, retrieval variant and k at the reference setting, averaged over six models and three seeds, with k increasing along each line. Grey is few-shot prompting, blue CICLe, hollow markers Fixed and filled markers Per-Class retrieval. Horizontal lines are the base classifier and fine-tuned RoBERTa-base. Ohsumed Per-Class is $k = 1$ only. Bootstrap intervals are in Table 2.

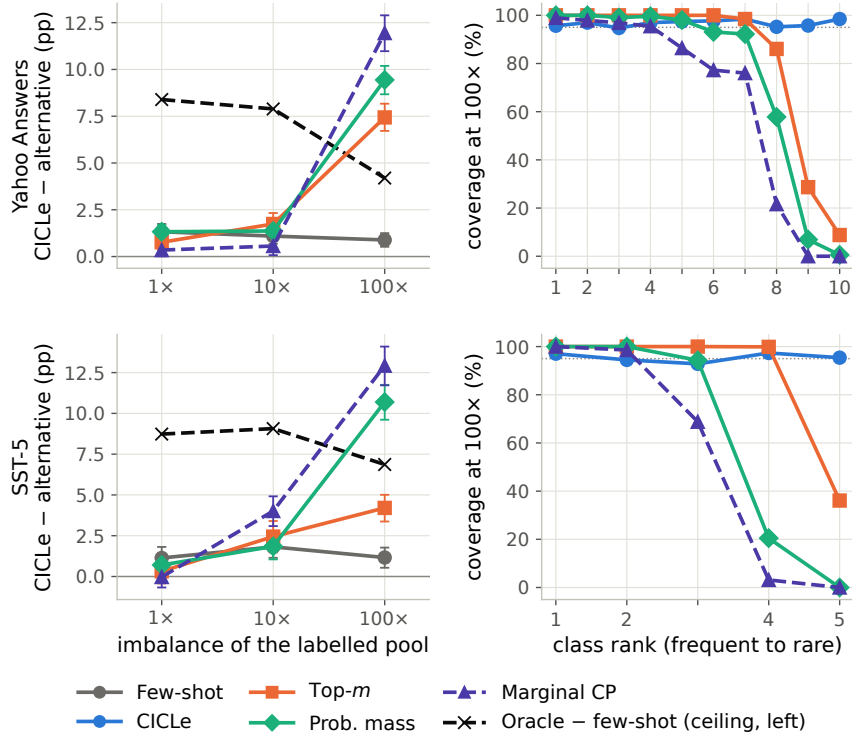


Figure 2: Narrowing rules under a long-tailed labelled pool on Yahoo Answers (top) and SST-5 (bottom), Per-Class retrieval. Left (a): CICLe minus few-shot, top- m , probability mass and marginal CP against the imbalance of the labelled pool, paired over six models, three seeds and $k \in \{1, 4\}$ (36 pairs per point) with 95% bootstrap intervals, and the oracle set minus few-shot as the dashed ceiling. Right (b): per-class coverage of the candidate sets at $100\times$ by class-frequency rank in the pool, mean over three seeds, with the $1 - \alpha$ line dotted. The test sample is the same at every imbalance level. Detailed version in Appendix D.

A. Macro-F1 at $k = 4$ (small numbers: mean prompt tokens per LLM call)

	Yahoo Answers	SST-5	SemEval-18	GoEmotions	Ohsumed
MiniLM + LR (no LLM)	65.4	35.5	9.7	11.2	45.2
RoBERTa-base, fine-tuned	61.8	51.2	24.5	37.0	55.4
Zero-shot	55.3 221	40.3 184	12.2 236	24.9 220	35.9 579
Candidate set only ($k=0$)	56.3 212	45.1 181	13.0 225	26.2 207	40.2 509
Few-shot, Fixed	58.4 455	44.1 298	15.3 338	26.4 305	50.4 1,826
CICLe, Fixed	58.2 447	45.8 295	15.9 328	26.8 293	51.7 1,756
Few-shot, Per-Class	61.6 2,166	46.6 746	15.4 2,209	24.8 2,422	44.6 ^{$k=1$} 7,518
CICLe, Per-Class	62.7 1,378	47.5 615	15.7 1,911	26.0 1,942	49.8 ^{$k=1$} 4,159

B. Paired gain of CICLe over few-shot prompting and prompt tokens saved

	Yahoo Answers	SST-5	SemEval-18	GoEmotions	Ohsumed
Fixed	-0.19 [-0.57, +0.20]	+1.77 [+1.14, +2.39]	+0.52 [+0.30, +0.73]	+0.59 [+0.19, +1.07]	+1.15 [+0.54, +1.87]
Per-Class	+1.40 [+1.04, +1.76]	+1.07 [+0.42, +1.71]	+0.46 [+0.22, +0.71]	+0.92 [+0.50, +1.38]	+5.21 [+4.43, +6.08]
Tokens saved	36%	18%	14%	20%	45% [†]

Table 2: Main results. LLM rows are means over six models and three seeds, supervised rows over three seeds. Δ is CICLe minus few-shot in macro-F1 points pooled over all k , with its 95% paired bootstrap interval over test instances and the number of (model, seed, k) cells. Bold marks intervals that exclude zero. Ohsumed Per-Class is run at $k = 1$ only. Tokens saved compare Per-Class CICLe with Per-Class few-shot prompting at equal k .

ference on SST-5 or SemEval-18 exceeds one point (Table 6, SST-5 and SemEval-18 rows). Under imbalance the conformal sets keep their coverage while the other rules lose it on the rare classes (Figure 2, right), and an LLM cannot recover a label it is not offered. At 100 times imbalance CICLe beats the alternatives by 4 to 13 points (Table 6, 100 $\times$ rows), the marginal rule being worst because its shared threshold is set by the confident head classes.

The naturally long-tailed datasets show the same effect at smaller size. On GoEmotions and Ohsumed CICLe is one to three points ahead of every alternative, and the lowest per-class coverage of every alternative there is zero (Table 7). Two things do not happen. CICLe’s margin over few-shot prompting does not grow with imbalance (Figure 2, left, grey line), because few-shot prompting lists every label and cannot lose coverage. And narrowing is not free of harm. When the set misses the gold label, which the $1 - \alpha$ coverage bounds, CICLe can only break an answer that few-shot prompting had right (Table 14, Appendix I). An oracle set that always contains the gold label shows that on balanced data the remaining room is in ranking, not in coverage (Table 6, oracle column). So far the LLM has seen label names it understands, which the next subsection removes.

5.4 Narrowing Helps Most with Uninformative Label Names

Narrowing helps most when the label names mean nothing. On all six datasets we replace every class name by a nonsense word, the same for all seeds, which leaves the base classifier untouched and forces the LLM to learn the labels from the examples (Table 8, Appendix E). With few examples few-shot prompting collapses, and CICLe’s advantage on Yahoo Answers grows to +6.5 [+6.0, +7.1] points with Fixed retrieval, the largest gain we measure (Table 8, Yahoo Answers rows). On SST-5 the advantage roughly doubles, and on the datasets whose names were weak hints to begin with it stays close to the original (Table 8). The candidate list without examples is useless under nonsense names (Table 8, $k = 0$ rows), so what CICLe adds here is examples from the right classes. A calibrated set also beats uncalibrated sets of the same size more clearly than under real names (Table 6, renamed rows), with one exception on renamed Ohsumed, where the much smaller marginal sets win by three points. Whether such gains justify an LLM at all is the next question.

5.5 When the LLM Pipeline Is Worth It

An LLM pipeline pays when the labelled pool is skewed or small, and a fine-tuned encoder wins when it is balanced and plentiful. Table 3 puts the fixed configuration, Per-Class CICLe at $k = 4$, next to the base classifier and two fine-tuned en-

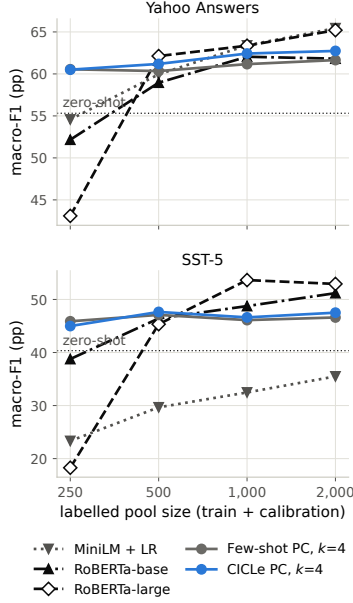


Figure 3: Macro-F1 against the labelled-pool size (log scale). Encoders: three seeds. Pipelines (Per-Class, $k = 4$): six models, three seeds. Dotted: zero-shot. Same test sample at every size.

	LR	RoB _B	RoB _L	FS	CICLe
Yahoo 1×	65.4	61.8	65.2	61.6	62.7
Yahoo 10×	50.4	58.7	62.9	59.4	60.3
Yahoo 100×	31.4	41.2	43.6	54.6	55.4
SST-5 1×	35.5	51.2	52.9	46.6	47.5
SST-5 10×	20.5	44.5	47.4	44.6	46.5
SST-5 100×	12.6	31.9	32.5	40.8	41.9
SemEval	9.7	24.5	23.2	15.4	15.7
GoEmo	11.2	37.0	34.3	24.8	26.0
Ohsumed	45.2	55.4	65.6	50.4 ^F	51.7 ^F

Table 3: Supervised baselines against the LLM pipelines, macro-F1, balanced and long-tailed pools of 1,600 examples. LR: MiniLM with logistic regression. RoB_B, RoB_L: fine-tuned RoBERTa-base and -large. FS, CICLe: Per-Class, $k = 4$ (F: Fixed). Bold: best per row. Full version in Table 10.

coders, and Figure 3 varies the size of the labelled pool. With 1,600 balanced labels RoBERTa-base beats every 3B to 8B pipeline on SST-5, SemEval-18, GoEmotions and Ohsumed, and on Yahoo Answers the logistic regression that CICLe narrows with beats both on its own (Table 3, 1× rows). The qualification is model size, since Qwen2.5-32B with CICLe overtakes the encoders on Yahoo Answers and SST-5 but not on the two many-class datasets (Table 13, Appendix H).

At 100 times imbalance the order flips. Both encoders lose about 20 points on Yahoo Answers while the pipeline loses seven (Table 3, 100× rows),

because retrieval and the LLM’s knowledge of the labels carry the rare classes. The pool-size sweep agrees (Figure 3). The pipelines are nearly flat between 250 and 2,000 labels, while RoBERTa-base climbs with the pool and overtakes them on SST-5 between 500 and 1,000 labels (Table 11, Appendix F). What remains is how tight the narrowing should be.

5.6 Choosing α and the Components

The miscoverage level should follow the strength of the base classifier. Figure 6 and Table 12 (Appendix G) vary α , the embedding and the classifier with two Llama models. Loosening α from 0.05 to 0.20 on Yahoo Answers, where the classifier is strong, turns a null difference into a gain of +3.6 [+2.7, +4.5] points while a third of the inputs never reach the LLM, and it also helps where the sets are otherwise large, at the price of the coverage guarantee. The gain survives a second prompt template and random retrieval (Table 9), and larger models gain less but do not lose (Table 13).

6 Conclusion

How a small classifier narrows the label space matters once the labelled pool is long-tailed. Class-conditional conformal sets keep the rare classes, top- m , probability-mass and marginal sets lose them, and the LLM cannot recover a label it was not offered. Rule: calibrate per class whenever the pool is skewed.

Narrowing gives a small, reliable gain over few-shot prompting and a shorter prompt, most when label names carry no meaning. Rule: expect a point or two and a third fewer tokens, and more when labels are codes.

An LLM pipeline pays when labels are skewed or scarce, and a fine-tuned encoder or the base classifier wins when they are plentiful and balanced. Rule: fine-tune on a large balanced pool, otherwise prompt with a narrowed list and loosen α as the classifier improves.

Prompts with different label information compare formats, not methods. Rule: list the admissible labels, or the candidate set for very large label spaces, in every prompt.

Limitations

Our study covers single-label classification in English, and GoEmotions is restricted to its single-label comments. Our label spaces have at most

59 classes, so listing every label costs little here, and for label spaces with hundreds or thousands of classes only the candidate set can be listed. CICLe is defined for a single label per input, so multi-label narrowing would need a different set construction. Every test sample holds 1,000 instances per seed. The paired bootstrap resolves differences of about half a point over the six-model grid, but per-model intervals (12 or 6 pairs) are two to three points wide, and the seed-42 checks of Section 5.6 are directional only. In the controlled-imbalance experiments the labelled pool is long-tailed but the test sample stays balanced, so the reported macro-F1 describes a deployment whose test distribution differs from its training distribution. A test sample with the same skew would weight the rare classes less. The LLM configurations are not selected on held-out data. We report one fixed configuration (Per-Class, $k = 4$, $\alpha = 0.05$) as the main number and state when a best configuration is chosen on the test set. The miscoverage level is likewise reported as an ablation, and the rule for choosing it is a heuristic. The conformal guarantee is per class and marginal over inputs within a class. It assumes that calibration and test data are exchangeable, which the pool-size and imbalance experiments respect by construction but a deployment may not. We checked one prompt family with one alternative wording and one example order, on one seed. Six open models with 3B to 8B parameters form the main grid, and the 12B and 32B models are run on the five short-text datasets only. Proprietary models and models above 32B parameters are not covered. Label renaming is an extreme stress test. Real deployments lie between informative and meaningless label names. The supervised baselines use one untuned fine-tuning recipe, which is unstable for RoBERTa-large on the smallest pools, so the encoder numbers are a floor rather than a ceiling for fine-tuning.

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A Protocol Details, Prompt Template and Output Parsing

This appendix gives the reasoning behind the sample sizes, the five protocol changes in detail, the prompt template, the parser and the original prompt format. It supports the label-information claim of Section 5.1 and makes every run reproducible.

Why these sizes. Two thousand labelled examples per dataset are enough for a usable base classifier on 5 to 28 classes and few enough that every configuration runs for six models and three seeds, at about one GPU-minute per Fixed configuration and up to 17 for Per-Class retrieval at $k = 8$ on a 24 GB GPU. The 400-example calibration split is the 20% of the original CICLE paper. The 1,000-instance test sample is shared by every method on a seed, and its bootstrap interval for a single run is one to two points of macro-F1. Three seeds change the subsample, the calibration split and the conformal tie-breaking. The Per-Class runs at $k = 8$ cost half of all compute, which is why every follow-up experiment uses $k \in \{1, 4\}$. Ohsumed abstracts average 270 tokens, so a Per-Class prompt at $k = 4$ with 23 classes would hold about 28,000 tokens, and Per-Class retrieval is run at $k = 1$ only (23 abstracts, 7,500 tokens). The 70B model of the Llama 3 family does not fit the available GPUs.

The five fixes in detail. (i) The prompt lists the admissible labels for every method, with the same wording for few-shot and CICLE prompts, so the candidate list and the retrieved examples are the only difference between conditions. (ii) The retrieval pool is the 1,600-example training split for every method. (iii) The subsample, the calibration split and the conformal sets (including the random tie-breaking of the smoothed p-values) are seeded, so that two methods on the same seed see the same test instances and the same candidate sets. (iv) The

generation budget is the token length of the longest label plus four, and outputs are normalised before matching. (v) Every run stores, per test instance, the gold label, the raw output, the parsed prediction, the candidate set, the retrieved example ids and the prompt length, from which every analysis in this paper is computed. Macro-F1 is computed over the classes present in the test sample, with invalid outputs coded as an extra class.

Prompt template. Every prompt is one user turn in the model’s chat template, built as follows ($\{\text{task}\}$ is one sentence per dataset, for example “We classify user questions into topic categories based on their text.”).

```
{task} The possible classes are:
{label1}, {label2}, . . . . Here are some
labelled examples:
"{example text}" => {label}
. . .
Please predict the correct class for the
following sample. Only provide the class
label.
"{input text}" =>
```

Zero-shot and few-shot prompts list all classes, narrowing methods list the candidate set, and $k = 0$ omits the example block. Examples are ordered from most to least similar. Labels that contain commas are separated by semicolons. Generation is greedy with a budget of the longest label’s token length plus four. The parser takes the first non-empty line of the output, applies NFKC normalisation, strips quotes, markdown markers and punctuation, and lower-cases it. An exact match to a label is accepted, then a label that is a prefix of the output followed by a word boundary (so that “Sports (football)” parses as Sports). Anything else is an invalid output and counts as wrong. The original prompt format of the released CICLE implementation, rerun in Appendix B, differs in three ways, and its missing label list is a design choice that fits Per-Class retrieval, where every candidate class appears among the examples, and saves context for label spaces with hundreds of classes. Its few-shot and CICLE prompts contain no label list and use different wording (“Here are some labelled examples sorted from most probable to least probable” for CICLE), generation is capped at five tokens, and the output must match a label exactly.

B Original Prompt Format Against the Corrected Protocol

Table 4 gives, per dataset and model on seed 42 with Fixed retrieval and $k \in \{1, 4\}$, the invalid-output rate and macro-F1 under the original prompt format and under the corrected protocol, together with the per-model paired difference between CICLE and few-shot prompting under both. It supports the label-information claim of Section 5.1: with Fixed retrieval, where only the retrieved labels appear, the invalid outputs concentrate in Mistral-3B, Qwen2.5-3B and Mistral-7B and in the datasets whose label names are not common words, and the sign of the per-model difference changes with the format in 11 of 30 cells.

C Per-Model Results

Table 5 breaks the main comparison down by model, with 12 paired cells per dataset and retrieval variant over three seeds. It supports the label-information claim of Section 5.1: under the corrected protocol Per-Class CICLE never significantly hurts any of the six models, and Fixed CICLE hurts two models on Yahoo Answers only, which is a smaller and different model dependence from the one the original prompt format produces in Table 4.

D Narrowing Comparisons

Table 6 gives the paired difference between CICLE and every alternative rule for all dataset variants and both retrieval variants, Table 7 the coverage, mean size, singleton rate and lowest per-class coverage of every rule’s candidate sets, Figure 4 the coverage against set size and the per-class F1 panels, and Figure 5 the main comparison for Fixed retrieval. They support the narrowing claim of Section 5.3: every rule covers 95 to 99% of gold labels overall, but only the class-conditional rule keeps the rare classes covered, and the macro-F1 gap under imbalance follows that coverage. The candidate sets do not depend on the LLM, k or the retrieval variant, so the set statistics come from one run per seed.

E Label Renaming and Prompt Robustness

Table 8 gives the label-renaming results for all six datasets, including the candidate set without examples, and Table 9 repeats the CICLE against few-shot comparison on seed 42 with a second prompt

template (“Choose exactly one of these labels”, examples as Text and Label lines, labels in random order and the most similar example placed last, which also tests the recency effect) and with examples drawn at random from the admissible classes. They support the claims of Sections 5.4 and 5.6: the gain from narrowing is largest when label names carry no meaning, and it does not depend on one prompt wording or on similarity-based retrieval. The nonsense words are listed in Appendix J.

F Supervised Baselines and Pool Size

Table 10 gives the supervised baselines against the LLM pipelines on every dataset and imbalance level, and Table 11 gives macro-F1 against the size of the labelled pool on the two balanced datasets. They support the claim of Section 5.5 about when the LLM pipeline is worth it: the pipelines are nearly flat across pool size and imbalance while the encoders and the base classifier are not. The base classifier, the retrieval pool and the fine-tuned encoders all see 80% of the labelled pool, and the calibration split is the remaining 20% (50 examples at a pool of 250), at which CICLE’s Per-Class margin over few-shot prompting on SST-5 is -0.63 $[-1.33, +0.13]$ and turns positive from 500 labels on.

G Ablations and the Miscoverage Level

Table 12 varies the embedding, the classifier and α with Llama-3.1-8B and Llama-3.2-3B on the five short-text datasets, and Figure 6 plots the α rows.

H Larger Models

Table 13 runs Mistral-NeMo-12B and Qwen2.5-32B with and without CICLE under the same protocol as the six small models, and lists the best 3B model with CICLE against the 32B model’s zero-shot score on the same seed. It supports the size qualification of Section 5.5 and the robustness statement of Section 5.6: the larger models gain less from narrowing but do not lose, and the 32B pipeline overtakes the fine-tuned encoders on Yahoo Answers and SST-5 only.

I Per-Instance Analyses

Table 14 pairs every test instance between CICLE and few-shot prompting at $k = 4$ and splits the fixes and breaks by whether the gold label was in the candidate set and by whether the few-shot answer was, Table 15 gives the invalid-output rate

Dataset	Model	invalid outputs (%): ZS / FS / CICLE		macro-F1: ZS / FS / CICLE		Δ CICLE — few-shot	
		original	corrected	original	corrected	original	corrected
Yahoo Answers	Llama-3.2-3B	0.6 / 2.9 / 4.6	1.9 / 0.6 / 0.5	51.0 / 54.5 / 55.6	50.6 / 49.4 / 45.6	+1.07 ^{n.s.} [-0.06, +2.19]	-3.79 [-5.58, -2.04]
	Ministral-3B	1.0 / 49.4 / 48.4	1.5 / 0.8 / 0.3	49.2 / 34.3 / 33.6	51.4 / 45.7 / 45.1	-0.76 ^{n.s.} [-2.36, +0.88]	-0.59 ^{n.s.} [-2.33, +1.19]
	Qwen2.5-3B	13.8 / 34.1 / 28.9	11.1 / 13.1 / 7.5	49.7 / 48.5 / 49.8	49.8 / 53.0 / 53.2	+1.34 [+0.14, +2.54]	+0.20 ^{n.s.} [-1.34, +1.78]
	Mistral-7B	20.8 / 27.9 / 28.8	10.1 / 5.6 / 3.0	55.4 / 55.0 / 55.9	56.4 / 57.6 / 56.8	+0.91 ^{n.s.} [-0.11, +1.92]	-0.88 ^{n.s.} [-2.24, +0.47]
	Qwen2.5-7B	8.8 / 16.0 / 15.0	10.9 / 4.0 / 2.2	56.2 / 57.1 / 56.9	55.3 / 61.8 / 60.2	-0.25 ^{n.s.} [-1.05, +0.55]	-1.67 [-2.76, -0.62]
	Llama-3.1-8B	0.3 / 8.4 / 7.3	0.3 / 0.2 / 0.1	57.8 / 58.4 / 58.9	58.7 / 60.4 / 61.0	+0.49 ^{n.s.} [-0.46, +1.45]	+0.66 ^{n.s.} [-0.51, +1.93]
SST-5	Llama-3.2-3B	0.0 / 0.0 / 0.1	0.0 / 0.1 / 0.1	32.0 / 40.0 / 41.3	35.7 / 36.6 / 39.5	+1.38 ^{n.s.} [-0.27, +3.01]	+2.88 [+0.86, +4.87]
	Ministral-3B	0.0 / 0.3 / 0.2	0.0 / 0.0 / 0.0	40.1 / 37.6 / 34.9	40.0 / 38.9 / 44.2	-2.64 [-4.51, -0.81]	+5.24 [+3.35, +7.18]
	Qwen2.5-3B	0.1 / 0.0 / 0.0	0.0 / 0.0 / 0.0	44.5 / 42.4 / 42.2	42.7 / 45.1 / 43.2	-0.22 ^{n.s.} [-1.58, +1.19]	-1.96 [-3.75, -0.03]
	Mistral-7B	4.7 / 8.4 / 5.0	0.1 / 0.0 / 0.0	40.6 / 44.2 / 46.3	42.5 / 44.8 / 44.5	+2.07 [+0.82, +3.38]	-0.29 ^{n.s.} [-1.72, +1.15]
	Qwen2.5-7B	0.0 / 2.5 / 2.2	0.0 / 0.0 / 0.0	39.0 / 46.8 / 48.1	39.8 / 47.4 / 48.2	+1.36 [+0.13, +2.63]	+0.78 ^{n.s.} [-0.65, +2.33]
	Llama-3.1-8B	0.0 / 1.4 / 1.1	0.0 / 0.0 / 0.0	31.8 / 46.0 / 47.1	39.9 / 42.8 / 45.0	+1.16 ^{n.s.} [-0.11, +2.45]	+2.24 [+0.42, +4.06]
SemEval-18	Llama-3.2-3B	5.2 / 33.7 / 31.2	0.1 / 2.3 / 2.8	9.3 / 9.8 / 9.3	11.1 / 13.7 / 13.9	-0.47 ^{n.s.} [-1.41, +0.47]	+0.17 ^{n.s.} [-1.00, +1.34]
	Ministral-3B	4.0 / 42.4 / 39.0	0.0 / 0.9 / 0.8	9.8 / 12.0 / 12.3	11.6 / 13.0 / 14.3	+0.34 ^{n.s.} [-0.43, +1.20]	+1.29 [+0.36, +2.24]
	Qwen2.5-3B	16.0 / 57.0 / 54.9	9.9 / 20.9 / 19.0	9.4 / 10.9 / 11.2	11.6 / 15.6 / 15.7	+0.28 ^{n.s.} [-0.40, +0.96]	+0.06 ^{n.s.} [-0.71, +0.86]
	Mistral-7B	31.3 / 62.8 / 57.5	7.9 / 7.8 / 7.1	9.8 / 10.3 / 10.8	11.1 / 15.4 / 17.0	+0.50 ^{n.s.} [-0.40, +1.39]	+1.65 [+0.77, +2.57]
	Qwen2.5-7B	14.0 / 36.8 / 33.1	13.4 / 8.2 / 7.9	13.0 / 12.3 / 12.9	13.1 / 16.3 / 16.7	+0.65 ^{n.s.} [-0.09, +1.44]	+0.36 ^{n.s.} [-0.57, +1.31]
	Llama-3.1-8B	44.3 / 11.8 / 10.6	1.4 / 1.0 / 0.8	11.3 / 12.1 / 12.4	15.2 / 16.9 / 16.9	+0.30 ^{n.s.} [-0.30, +0.91]	+0.02 ^{n.s.} [-0.88, +0.94]
GoEmotions	Llama-3.2-3B	0.3 / 4.2 / 4.8	0.2 / 1.0 / 0.7	15.1 / 14.1 / 14.9	15.9 / 17.5 / 17.8	+0.82 [+0.19, +1.50]	+0.33 ^{n.s.} [-0.88, +1.53]
	Ministral-3B	0.4 / 46.7 / 42.3	0.1 / 0.2 / 0.1	25.1 / 15.2 / 14.7	24.0 / 22.4 / 21.4	-0.53 ^{n.s.} [-1.57, +0.57]	-0.96 ^{n.s.} [-2.86, +1.24]
	Qwen2.5-3B	5.1 / 29.5 / 27.5	4.8 / 6.2 / 5.5	25.0 / 19.0 / 19.7	27.0 / 26.8 / 27.0	+0.66 ^{n.s.} [-0.37, +1.73]	+0.28 ^{n.s.} [-1.14, +1.91]
	Mistral-7B	23.8 / 35.4 / 32.4	7.5 / 7.7 / 5.6	25.1 / 20.6 / 19.9	28.7 / 30.4 / 30.9	-0.65 ^{n.s.} [-1.90, +0.68]	+0.47 ^{n.s.} [-1.16, +2.18]
	Qwen2.5-7B	4.5 / 22.6 / 20.4	4.5 / 5.4 / 5.0	32.8 / 20.2 / 21.5	32.5 / 33.1 / 32.9	+1.33 [+0.15, +2.63]	-0.20 ^{n.s.} [-1.92, +1.63]
	Llama-3.1-8B	1.3 / 14.5 / 9.7	1.0 / 1.0 / 0.8	23.3 / 17.9 / 18.2	23.9 / 25.7 / 25.5	+0.27 ^{n.s.} [-0.77, +1.37]	-0.17 ^{n.s.} [-1.71, +1.49]
Ohsumed	Llama-3.2-3B	14.9 / 33.1 / 35.7	2.7 / 0.8 / 0.0	16.2 / 30.1 / 30.9	27.3 / 50.0 / 49.3	+0.78 ^{n.s.} [-0.18, +1.63]	-0.73 ^{n.s.} [-2.88, +1.42]
	Ministral-3B	34.5 / 65.2 / 64.4	0.7 / 0.8 / 0.1	24.4 / 20.5 / 20.5	37.9 / 36.7 / 39.2	+0.01 ^{n.s.} [-1.40, +1.46]	+2.54 [+0.16, +5.00]
	Qwen2.5-3B	55.6 / 55.5 / 52.7	21.7 / 19.2 / 6.9	14.5 / 26.0 / 27.6	26.3 / 40.9 / 45.5	+1.63 [+0.48, +2.68]	+4.58 [+2.17, +7.11]
	Mistral-7B	64.3 / 62.8 / 63.6	2.3 / 3.7 / 0.6	11.4 / 16.1 / 16.3	39.5 / 48.1 / 48.5	+0.17 ^{n.s.} [-0.36, +0.73]	+0.47 ^{n.s.} [-1.64, +2.62]
	Qwen2.5-7B	31.0 / 39.4 / 39.2	2.8 / 2.5 / 0.9	23.5 / 33.1 / 32.9	41.3 / 59.6 / 59.8	-0.17 ^{n.s.} [-1.60, +0.72]	+0.22 ^{n.s.} [-1.89, +2.09]
	Llama-3.1-8B	32.2 / 36.8 / 37.2	1.0 / 0.2 / 0.0	24.4 / 32.2 / 32.4	46.2 / 61.3 / 60.6	+0.12 ^{n.s.} [-0.79, +0.82]	-0.69 ^{n.s.} [-2.57, +1.03]

Table 4: The original prompt format of the released CICLE implementation against the corrected protocol on the same seed-42 test sample, per dataset and model, Fixed retrieval, $k \in \{1, 4\}$. Invalid-output rate and macro-F1 for zero-shot (ZS), few-shot (FS) and CICLE, the latter two averaged over k . Original = no label list in the few-shot and CICLE prompts (a choice suited to Per-Class retrieval, not run here), different wording, five-token generation cap, exact matching (column heading “original”). Δ = CICLE minus few-shot paired over the 1,000 instances and the two values of k , with 95% bootstrap intervals.

Model	Yahoo Answers F	Yahoo Answers PC	SST-5 F	SST-5 PC	SemEval-18 F	SemEval-18 PC	GoEmotions F	GoEmotions PC	Ohsumed F
Llama-3.2-3B	-2.33 [-3.13, -1.53]	+0.82 [+0.22, +1.42]	+3.85 [+2.80, +4.92]	+3.17 [+2.14, +4.24]	+0.50 [+0.02, +0.96]	+0.92 [+0.41, +1.43]	+0.92 [+0.26, +1.62]	+2.46 [+1.39, +3.47]	-0.33 ^{n.s.} [-1.53, +0.98]
Ministral-3B	-0.17 ^{n.s.} [-1.09, +0.77]	+3.40 [+2.55, +4.25]	+3.36 [+2.31, +4.43]	+1.33 [+0.33, +2.36]	+0.74 [+0.24, +1.23]	+0.62 [+0.13, +1.11]	+0.47 ^{n.s.} [-0.40, +1.55]	+1.32 [+0.49, +2.14]	+1.66 [+0.33, +3.08]
Qwen2.5-3B	+1.99 [+1.23, +2.79]	+2.14 [+1.35, +2.92]	+0.40 ^{n.s.} [-0.60, +1.40]	+1.12 [+0.03, +2.23]	+0.12 ^{n.s.} [-0.28, +0.49]	+0.16 ^{n.s.} [-0.31, +0.60]	+0.39 ^{n.s.} [-0.26, +1.11]	+0.95 [+0.19, +1.75]	+4.71 [+3.20, +6.65]
Mistral-7B	-0.69 [-1.31, -0.06]	+0.08 ^{n.s.} [-0.58, +0.75]	-0.42 ^{n.s.} [-1.27, +0.47]	-0.54 ^{n.s.} [-1.48, +0.42]	+0.85 [+0.39, +1.31]	+0.73 [+0.22, +1.23]	+0.73 [+0.03, +1.40]	+0.15 ^{n.s.} [-0.63, +1.10]	-0.13 ^{n.s.} [-1.32, +1.12]
Qwen2.5-7B	+0.05 ^{n.s.} [-0.44, +0.54]	+1.24 [+0.67, +1.84]	+1.49 [+0.64, +2.35]	+0.96 [+0.07, +1.85]	+0.48 [+0.05, +0.93]	-0.29 ^{n.s.} [-0.76, +0.21]	+0.69 ^{n.s.} [-0.35, +2.02]	+0.45 ^{n.s.} [-0.59, +1.50]	+0.49 ^{n.s.} [-0.62, +1.61]
Llama-3.1-8B	+0.03 ^{n.s.} [-0.49, +0.57]	+0.74 [+0.20, +1.30]	+1.94 [+1.06, +2.81]	+0.35 ^{n.s.} [-0.59, +1.28]	+0.45 ^{n.s.} [-0.01, +0.90]	+0.65 [+0.14, +1.16]	+0.32 ^{n.s.} [-0.52, +1.18]	+0.20 ^{n.s.} [-0.76, +1.20]	+0.51 ^{n.s.} [-0.65, +1.61]

Table 5: CICLE minus few-shot prompting per model, paired over all k and three seeds (12 pairs, Ohsumed 6), with 95% bootstrap intervals over test instances. F = Fixed, PC = Per-Class. Bold marks intervals that exclude zero.

per model and dataset, Table 16 the most frequent unparseable outputs (emoji replaced by “[emoji]” for typesetting), and Table 17 qualitative examples of instances that narrowing fixed or broke. They support the narrowing claim of Section 5.3, since the breaks concentrate on instances whose gold label the set missed, and the label-information claim of Section 5.1, since the unparseable outputs are labels of the wrong granularity, emoji outside the label set and refusals rather than failures of the method.

J Dataset Details

MASSIVE uses the English intent split SetFit/amazon_massive_intent_en-US. One intent (cooking_query) has four training exam-

ples, too few to appear in both the training and calibration splits of a 2,000-example subsample, and is dropped, leaving 59 intents. The official test split holds a single example of one intent, too few to stratify, so the validation and test splits together form the test pool. The 2,000-example subsample keeps the natural skew, with 70 times as many examples in the largest class as in the smallest. Table 18 lists the class order and training-pool counts under controlled imbalance for every seed, and Table 19 the nonsense-word mapping used for label renaming. They document the two controlled variants behind the narrowing claim of Sections 5.3 and 5.4. Class sizes follow a geometric schedule between the largest and the smallest class, with at least five examples per class before the 80/20

Variant	Retrieval	vs. few-shot	vs. top- m	vs. prob. mass	vs. marginal CP	vs. oracle
Yahoo Answers	Fixed	-0.19 ^{n.s.} [-0.57, +0.20] (72)	+0.37 ^{n.s.} [-0.06, +0.79] (36)	+0.36 ^{n.s.} [-0.05, +0.77] (36)	-0.02 ^{n.s.} [-0.37, +0.33] (36)	-7.46 [-8.21, -6.74] (36)
Yahoo Answers	Per-Class	+1.40 [+1.04, +1.76] (72)	+0.76 [+0.35, +1.19] (36)	+1.33 [+0.94, +1.70] (36)	+0.35 [+0.02, +0.68] (36)	-7.07 [-7.83, -6.33] (36)
Yahoo Answers 10×	Fixed	-0.45 [-0.90, -0.03] (36)	+0.58 [+0.04, +1.13] (36)	+0.29 ^{n.s.} [-0.14, +0.73] (36)	-0.31 ^{n.s.} [-0.84, +0.22] (36)	-6.85 [-7.54, -6.19] (36)
Yahoo Answers 10×	Per-Class	+1.09 [+0.68, +1.50] (36)	+1.74 [+1.18, +2.33] (36)	+1.37 [+0.93, +1.81] (36)	+0.56 [+0.07, +1.07] (36)	-6.80 [-7.53, -6.12] (36)
Yahoo Answers 100×	Fixed	+0.21 ^{n.s.} [-0.16, +0.60] (36)	+6.52 [+5.79, +7.25] (36)	+9.08 [+8.30, +9.83] (36)	+10.81 [+9.81, +11.75] (36)	-3.10 [-3.69, -2.52] (36)
Yahoo Answers 100×	Per-Class	+0.89 [+0.52, +1.25] (36)	+7.44 [+6.72, +8.17] (36)	+9.44 [+8.67, +10.19] (36)	+11.94 [+10.98, +12.89] (36)	-3.31 [-3.86, -2.75] (36)
Yahoo Answers (renamed)	Fixed	+6.52 [+5.96, +7.09] (36)	+2.61 [+2.02, +3.18] (36)	+5.03 [+4.49, +5.58] (36)	—	—
Yahoo Answers (renamed)	Per-Class	+5.62 [+5.00, +6.25] (36)	+2.64 [+2.06, +3.22] (36)	+4.46 [+3.91, +5.03] (36)	—	—
SST-5	Fixed	+1.77 [+1.14, +2.39] (72)	-0.14 ^{n.s.} [-0.89, +0.61] (36)	+1.07 [+0.43, +1.70] (36)	-0.56 ^{n.s.} [-1.18, +0.04] (36)	-7.11 [-8.06, -6.11] (36)
SST-5	Per-Class	+1.07 [+0.42, +1.71] (72)	+0.32 ^{n.s.} [-0.42, +1.09] (36)	+0.71 [+0.08, +1.31] (36)	-0.03 ^{n.s.} [-0.67, +0.62] (36)	-7.60 [-8.53, -6.61] (36)
SST-5 10×	Fixed	+2.33 [+1.69, +2.99] (36)	+1.90 [+0.96, +2.82] (36)	+1.69 [+0.93, +2.43] (36)	+3.22 [+2.39, +4.08] (36)	-6.59 [-7.56, -5.61] (36)
SST-5 10×	Per-Class	+1.82 [+1.15, +2.48] (36)	+2.44 [+1.48, +3.40] (36)	+1.85 [+1.05, +2.61] (36)	+4.01 [+3.09, +4.91] (36)	-7.25 [-8.24, -6.22] (36)
SST-5 100×	Fixed	+1.51 [+0.87, +2.14] (36)	+3.35 [+2.57, +4.10] (36)	+8.54 [+7.54, +9.50] (36)	+10.28 [+9.15, +11.36] (36)	-5.33 [-6.20, -4.43] (36)
SST-5 100×	Per-Class	+1.16 [+0.54, +1.77] (36)	+4.21 [+3.37, +5.01] (36)	+10.70 [+9.61, +11.75] (36)	+12.92 [+11.72, +14.10] (36)	-5.70 [-6.58, -4.82] (36)
SST-5 (renamed)	Fixed	+3.75 [+3.18, +4.29] (36)	+1.96 [+1.40, +2.53] (36)	+2.63 [+2.13, +3.11] (36)	—	—
SST-5 (renamed)	Per-Class	+2.10 [+1.46, +2.73] (36)	-0.17 ^{n.s.} [-0.85, +0.51] (36)	+1.00 [+0.44, +1.55] (36)	—	—
SemEval-18	Fixed	+0.52 [+0.30, +0.73] (72)	+0.04 ^{n.s.} [-0.24, +0.32] (36)	+0.01 ^{n.s.} [-0.28, +0.30] (36)	—	-1.13 [-1.51, -0.74] (36)
SemEval-18	Per-Class	+0.46 [+0.22, +0.71] (72)	+0.01 ^{n.s.} [-0.30, +0.33] (36)	+0.02 ^{n.s.} [-0.29, +0.35] (36)	—	-1.32 [-1.68, -0.94] (36)
SemEval-18 (renamed)	Fixed	+0.79 [+0.54, +1.03] (36)	+0.14 ^{n.s.} [-0.12, +0.41] (36)	+0.21 ^{n.s.} [-0.06, +0.47] (36)	—	—
SemEval-18 (renamed)	Per-Class	+0.72 [+0.44, +1.00] (36)	+0.07 ^{n.s.} [-0.23, +0.36] (36)	+0.08 ^{n.s.} [-0.20, +0.37] (36)	—	—
GoEmotions	Fixed	+0.59 [+0.19, +1.07] (72)	+2.51 [+1.49, +3.23] (36)	+2.26 [+1.24, +2.98] (36)	+1.92 [+0.79, +2.73] (36)	-2.49 [-3.12, -1.73] (36)
GoEmotions	Per-Class	+0.92 [+0.50, +1.38] (72)	+2.54 [+1.31, +3.42] (36)	+1.91 [+0.72, +2.77] (36)	+1.28 [+0.02, +2.24] (36)	-2.35 [-2.98, -1.68] (36)
GoEmotions (renamed)	Fixed	+0.66 [+0.34, +1.00] (36)	+0.43 ^{n.s.} [-0.07, +0.89] (36)	+0.42 ^{n.s.} [-0.09, +0.90] (36)	—	—
GoEmotions (renamed)	Per-Class	+0.98 [+0.66, +1.39] (36)	+0.65 [+0.12, +1.19] (36)	+0.34 ^{n.s.} [-0.19, +0.94] (36)	—	—
Ohsumed	Fixed	+1.15 [+0.54, +1.87] (36)	+1.08 [+0.34, +1.80] (36)	+1.52 [+0.82, +2.24] (36)	+2.88 [+1.86, +4.08] (36)	-8.43 [-9.44, -7.40] (36)
Ohsumed	Per-Class	+5.21 [+4.43, +6.08] (18)	—	—	—	—
Ohsumed (renamed)	Fixed	+1.00 [+0.41, +1.79] (36)	+0.06 ^{n.s.} [-0.68, +0.94] (36)	+1.03 [+0.35, +1.80] (36)	-3.33 [-4.21, -2.05] (36)	-4.17 [-4.91, -3.21] (36)

Table 6: CICLE minus each alternative narrowing rule and few-shot prompting, paired macro-F1 difference with 95% bootstrap interval over test instances and the number of (model, seed, k) cells in parentheses. The few-shot column pools all k , the other columns $k \in \{1, 4\}$. Six small models, three seeds. Bold marks intervals that exclude zero.

split, and the class order is a random permutation drawn from the seed, so the rare classes differ between seeds and the per-class panels of Figure 2 are aggregated by frequency rank. The nonsense words are six-letter consonant-vowel strings generated with a fixed seed, no two sharing a four-letter prefix, so that the prefix matching of the parser cannot confuse them.

K Full Result Grids

Tables 20 to 33 give every method, retrieval variant and k for each dataset variant at the reference setting, with accuracy, invalid-output rate, mean number of examples and mean prompt tokens per LLM call, as means over the runs of the six small models. They support the gain claim of Section 5.2, in particular that the gain does not grow with k , and they are the source of every mean in the main text.

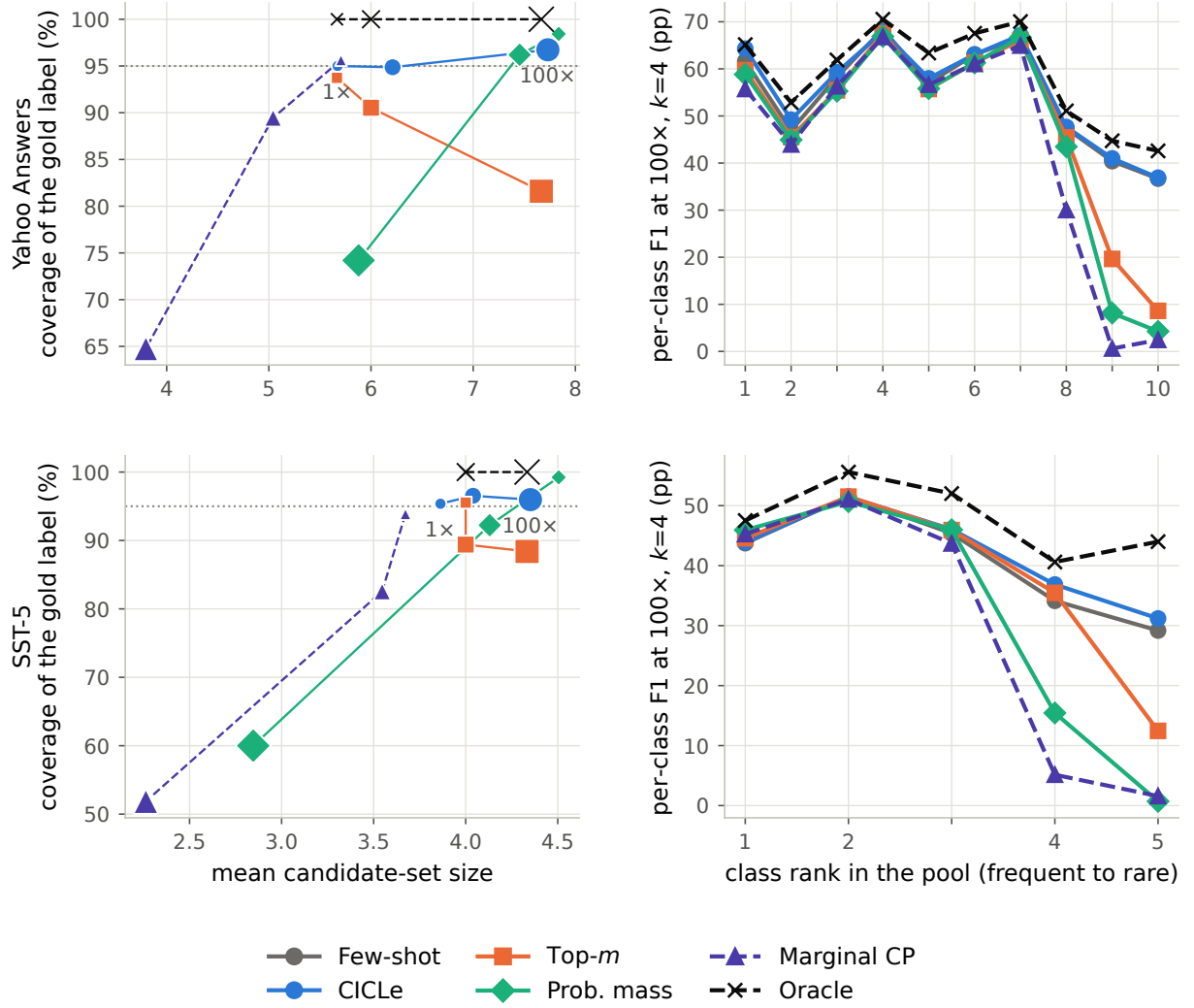


Figure 4: Further panels of the narrowing analysis (Per-Class retrieval, rows Yahoo Answers and SST-5). Left: coverage of the gold label against mean candidate-set size for every rule at 1, 10 and 100 times imbalance (marker size grows with imbalance), mean over three seeds, with the $1 - \alpha$ target dotted. Right: per-class F1 at 100 times imbalance and $k = 4$ by class-frequency rank in the pool, mean over six models and three seeds.

Variant	Method	coverage (%)	mean size	singleton (%)	min. per-class coverage (%)
Yahoo Answers	CICLe	95.0	5.67	3.4	90.6
	Top- m	93.7	5.67	0.0	87.1
	Prob. mass	98.4	7.84	0.1	96.7
	Marginal CP	95.6	5.71	5.1	91.0
	Oracle	100.0	5.67	0.0	100.0
Yahoo Answers 10×	CICLe	94.9	6.21	0.6	84.1
	Top- m	90.5	6.00	0.0	65.0
	Prob. mass	96.2	7.46	0.5	86.6
	Marginal CP	89.5	5.04	6.9	68.8
	Oracle	100.0	6.00	0.0	100.0
Yahoo Answers 100×	CICLe	96.7	7.73	0.0	92.2
	Top- m	81.6	7.67	0.0	8.8
	Prob. mass	74.2	5.88	1.8	0.6
	Marginal CP	64.8	3.80	10.9	0.0
	Oracle	100.0	7.67	0.0	100.0
Yahoo Answers (renamed)	CICLe	94.8	5.68	3.4	89.6
	Top- m	93.7	5.67	0.0	87.1
	Prob. mass	98.4	7.84	0.1	96.7
	Marginal CP	95.6	5.71	5.1	91.0
	Oracle	100.0	5.67	0.0	100.0
SST-5	CICLe	95.4	3.86	0.0	92.3
	Top- m	95.5	4.00	0.0	84.7
	Prob. mass	99.2	4.50	0.0	97.3
	Marginal CP	93.9	3.67	0.1	82.0
	Oracle	100.0	4.00	0.0	100.0
SST-5 10×	CICLe	96.5	4.04	0.7	91.5
	Top- m	89.4	4.00	0.0	56.6
	Prob. mass	92.2	4.13	0.4	70.2
	Marginal CP	82.6	3.55	4.0	42.1
	Oracle	100.0	4.00	0.0	100.0
SST-5 100×	CICLe	96.0	4.35	0.1	89.1
	Top- m	88.4	4.33	0.0	36.1
	Prob. mass	60.0	2.85	2.3	0.0
	Marginal CP	51.9	2.26	11.3	0.0
	Oracle	100.0	4.33	0.0	100.0
SST-5 (renamed)	CICLe	95.4	3.87	0.0	92.3
	Top- m	95.5	4.00	0.0	84.7
	Prob. mass	99.2	4.50	0.0	97.3
	Marginal CP	93.9	3.67	0.1	82.0
	Oracle	100.0	4.00	0.0	100.0
SemEval-18	CICLe	94.4	17.12	0.0	79.9
	Top- m	97.1	17.00	0.0	77.7
	Prob. mass	97.2	16.87	0.0	80.8
	Oracle	100.0	17.00	0.0	100.0
SemEval-18 (renamed)	CICLe	94.4	17.16	0.0	81.1
	Top- m	97.1	17.00	0.0	77.7
	Prob. mass	97.2	16.87	0.0	80.8
	Marginal CP	96.2	16.02	0.0	73.5
	Oracle	100.0	17.00	0.0	100.0
GoEmotions	CICLe	95.4	22.10	0.0	73.9
	Top- m	98.8	22.00	0.0	0.0
	Prob. mass	97.9	18.80	0.0	0.0
	Marginal CP	95.1	14.35	0.1	0.0
	Oracle	100.0	22.00	0.0	100.0
GoEmotions (renamed)	CICLe	95.5	22.13	0.0	75.6
	Top- m	98.8	22.00	0.0	0.0
	Prob. mass	97.9	18.80	0.0	0.0
	Marginal CP	95.2	14.34	0.1	0.0
	Oracle	100.0	22.00	0.0	100.0
Ohsumed	CICLe	96.4	12.27	0.0	84.9
	Top- m	98.8	12.00	0.0	0.0
	Prob. mass	99.4	13.95	0.5	0.0
	Marginal CP	95.2	4.81	7.4	0.0
	Oracle	100.0	12.00	0.0	100.0
Ohsumed (renamed)	CICLe	96.3 ¹⁷	12.28	0.0	76.8
	Top- m	98.8	12.00	0.0	0.0
	Prob. mass	99.4	13.95	0.5	0.0
	Marginal CP	95.2	4.81	7.4	0.0
	Oracle	100.0	12.00	0.0	100.0

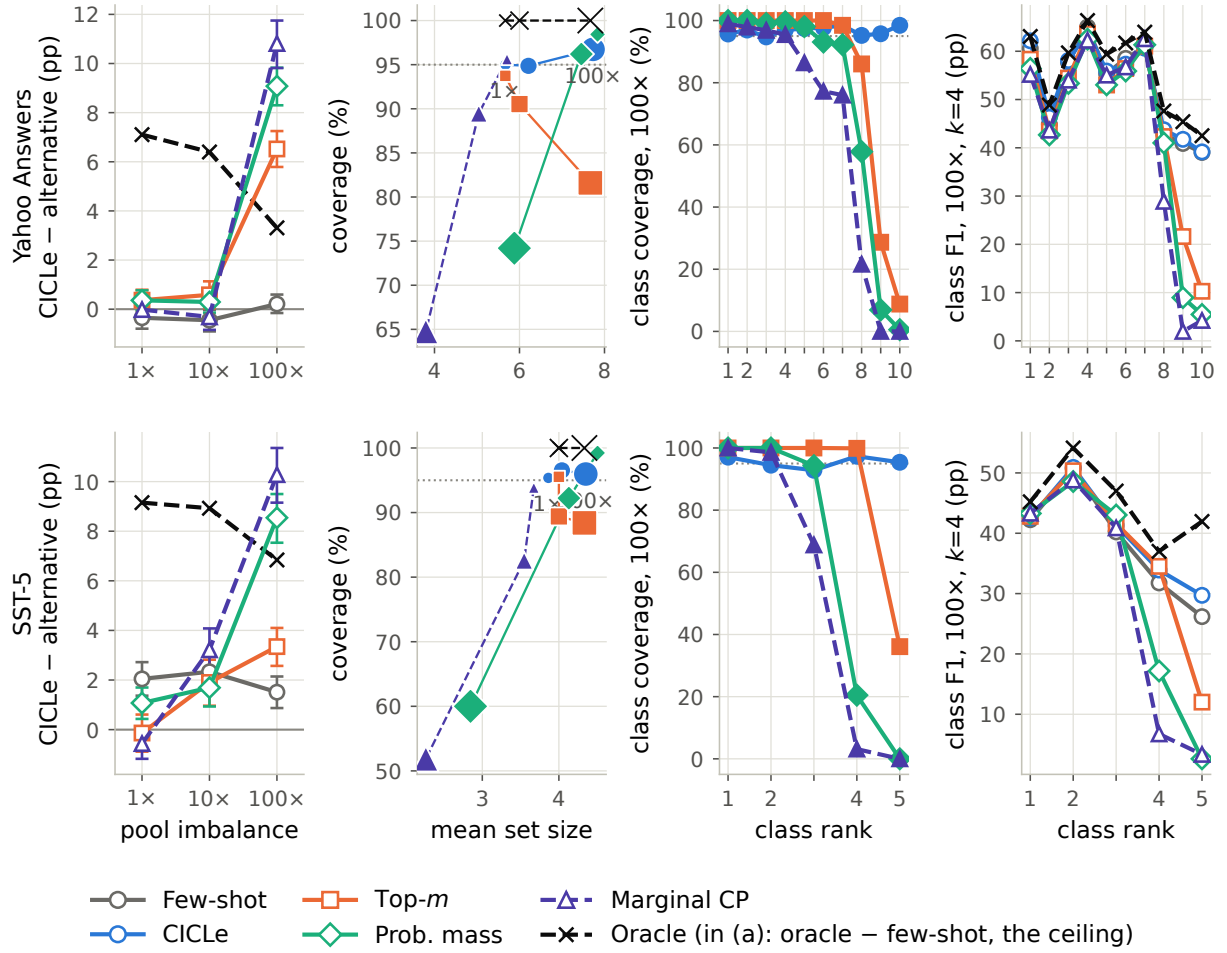


Figure 5: Figure 2 for the Fixed retrieval variant, with the same aggregation (36 pairs per point on the left, one run per seed on the right).

Dataset	Labels	k	FS Fixed	CICLe Fixed	FS PC	CICLe PC	Δ Fixed	Δ PC
Yahoo Answers	original	0	—	56.3	—	—		
Yahoo Answers	original	1	54.8	54.2	57.9	59.5	-0.35 ^{n.s.} [-0.79, +0.10]	+1.32 [+0.90, +1.74]
Yahoo Answers	original	4	58.4	58.2	61.6	62.7		
Yahoo Answers	nonsense words	0	—	18.9	—	—		
Yahoo Answers	nonsense words	1	25.3	34.3	34.1	41.1	+6.52 [+5.96, +7.09]	+5.62 [+5.00, +6.25]
Yahoo Answers	nonsense words	4	40.6	44.6	47.8	52.0		
SST-5	original	0	—	45.1	—	—		
SST-5	original	1	40.6	43.0	44.9	46.2	+2.05 [+1.37, +2.72]	+1.14 [+0.47, +1.81]
SST-5	original	4	44.1	45.8	46.6	47.5		
SST-5	nonsense words	1	24.3	28.4	33.3	35.4	+3.75 [+3.18, +4.29]	+2.10 [+1.46, +2.73]
SST-5	nonsense words	4	32.1	35.5	36.8	38.8		
SemEval-18	original	0	—	13.0	—	—		
SemEval-18	original	1	14.3	14.7	15.2	15.7	+0.49 [+0.23, +0.74]	+0.46 [+0.18, +0.74]
SemEval-18	original	4	15.3	15.9	15.4	15.7		
SemEval-18	nonsense words	1	6.3	7.0	7.7	8.3	+0.79 [+0.54, +1.03]	+0.72 [+0.44, +1.00]
SemEval-18	nonsense words	4	8.3	9.2	10.5	11.3		
GoEmotions	original	0	—	26.2	—	—		
GoEmotions	original	1	24.6	25.5	25.0	26.0	+0.62 [+0.17, +1.17]	+1.10 [+0.56, +1.65]
GoEmotions	original	4	26.4	26.8	24.8	26.0		
GoEmotions	nonsense words	0	—	2.1	—	—		
GoEmotions	nonsense words	1	8.0	8.8	10.6	11.7	+0.66 [+0.34, +1.00]	+0.98 [+0.66, +1.39]
GoEmotions	nonsense words	4	11.8	12.3	14.4	15.3		
Ohsumed	original	0	—	40.2	—	—		
Ohsumed	original	1	46.7	47.6	—	—	+1.15 [+0.54, +1.87]	
Ohsumed	original	4	50.4	51.7	—	—		
Ohsumed	nonsense words	0	—	4.5	—	—		
Ohsumed	nonsense words	1	27.1	27.8	—	—	+1.00 [+0.41, +1.79]	
Ohsumed	nonsense words	4	40.2	41.5	—	—		

Table 8: Label renaming. Macro-F1 with the original class names and with every name replaced by a nonsense word (same mapping for all seeds), mean over six models and three seeds. $k = 0$ is the candidate set without examples. $\Delta = \text{CICLe}$ minus few-shot, paired over $k \in \{1, 4\}$ (36 pairs), with 95% bootstrap intervals over test instances. The base classifier is unaffected by renaming, so the whole effect is on the LLM’s use of the candidate list and the examples. FS = few-shot, PC = Per-Class. Ohsumed is Fixed only.

Dataset	Condition	FS Fixed	CICLe Fixed	Δ Fixed	FS PC	CICLe PC	Δ PC
Yahoo Answers	default prompt	54.7	53.7	-1.01 [-1.77, -0.22]	57.6	59.0	+1.32 [+0.60, +2.03]
	alternative prompt	56.0	54.4	-1.65 [-2.48, -0.85]	53.8	53.6	-0.18 ^{n.s.} [-1.10, +0.71]
	random retrieval	52.8	52.6	-0.28 ^{n.s.} [-1.48, +0.86]	52.3	54.5	+2.26 [+1.24, +3.29]
SST-5	default prompt	42.6	44.1	+1.48 [+0.35, +2.59]	46.5	47.2	+0.69 ^{n.s.} [-0.39, +1.84]
	alternative prompt	44.3	45.8	+1.51 [+0.37, +2.72]	45.6	47.4	+1.88 [+0.78, +3.00]
	random retrieval	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]
SemEval-18	default prompt	15.1	15.7	+0.59 [+0.15, +1.04]	15.5	16.2	+0.68 [+0.24, +1.12]
	alternative prompt	15.2	15.5	+0.32 ^{n.s.} [-0.26, +0.89]	14.4	15.1	+0.71 [+0.10, +1.34]
	random retrieval	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]
GoEmotions	default prompt	26.0	25.9	-0.04 ^{n.s.} [-0.79, +0.85]	25.4	26.1	+0.77 ^{n.s.} [-0.01, +1.63]
	alternative prompt	25.6	26.0	+0.39 ^{n.s.} [-0.69, +1.54]	25.5	25.8	+0.32 ^{n.s.} [-0.99, +1.57]
	random retrieval	24.9	25.4	+0.52 ^{n.s.} [-0.42, +1.63]	23.5	23.9	+0.39 ^{n.s.} [-0.59, +1.71]
Ohsumed	default prompt	49.4	50.5	+1.07 ^{n.s.} [-0.06, +2.23]	—	—	—
	alternative prompt	46.0	47.3	+1.32 [+0.11, +2.50]	—	—	—
	random retrieval	[pending]	[pending]	[pending]	[pending]	[pending]	[pending]

Table 9: Prompt robustness on seed 42. CICLe against few-shot prompting under the default prompt, an alternative prompt wording and random instead of similarity-based example retrieval, six small models, $k \in \{1, 4\}$. Cells are means over the 12 runs, and Δ is paired over the 1,000 instances and 12 cells with its 95% bootstrap interval. Ohsumed is Fixed only.

	Yahoo Answers 1×	Yahoo Answers 10×	Yahoo Answers 100×	SST-5 1×	SST-5 10×	SST-5 100×	SemEval-18	GoEmotions	Ohsumed
MiniLM + LR	65.4	50.4	31.4	35.5	20.5	12.6	9.7	11.2	45.2
RoBERTa-base, fine-tuned	61.8	58.7	41.2	51.2	44.5	31.9	24.5	37.0	55.4
RoBERTa-large, fine-tuned	64.6	61.6	44.9	39.9	35.8	32.2	23.0	27.3	66.1
Zero-shot	55.3	—	—	40.3	—	—	12.2	24.9	35.9
Few-shot PC, $k=4$	61.6	59.4	54.6	46.6	44.6	40.8	15.4	24.8	50.4 ^F
CICLe PC, $k=4$	62.7	60.3	55.4	47.5	46.5	41.9	15.7	26.0	51.7 ^F
Best LLM pipeline	63.8 CICLe PC $k=8$	60.3 CICLe PC $k=4$	56.0 CICLe PC $k=1$	47.6 CICLe PC $k=2$	46.5 CICLe PC $k=4$	43.4 CICLe PC $k=1$	16.1 CICLe F $k=8$	27.3 CICLe F $k=8$	53.1 CICLe F $k=8$

Table 10: Supervised baselines against LLM pipelines, macro-F1. Supervised rows are means over three seeds, LLM rows over six models and three seeds. The fixed configuration is Per-Class retrieval at $k = 4$ (Ohsumed: Fixed, marked F). The best pipeline is the best (method, variant, k) mean per column and is selected on the test set. Zero-shot prompting does not depend on the pool. The test sample is identical at every imbalance level. Bold marks the best value per column.

Dataset	labelled pool	MiniLM + LR	RoBERTa-base	RoBERTa-large	Few-shot PC, $k=4$	CICLe PC, $k=4$	best LLM pipeline
Yahoo Answers	250	54.6	52.2	54.3	60.6	60.5	60.6 FS PC $k=4$
	500	60.0	59.0	43.1	60.3	61.2	61.2 CICLe PC $k=4$
	1,000	63.4	62.0	62.9	61.2	62.4	62.4 CICLe PC $k=4$
	2,000	65.4	61.8	64.6	61.6	62.7	63.8 CICLe PC $k=8$
SST-5	250	23.3	38.8	31.5	45.9	45.0	45.9 FS PC $k=4$
	500	29.7	46.4	36.2	47.1	47.6	47.6 CICLe PC $k=4$
	1,000	32.5	48.8	39.7	46.1	46.6	46.6 CICLe PC $k=4$
	2,000	35.5	51.2	39.9	46.6	47.5	47.6 CICLe PC $k=2$

Table 11: Macro-F1 against the size of the labelled pool (training plus calibration, split 80/20). Supervised rows are means over three seeds, LLM rows over six models and three seeds, and the test sample is the same at every pool size. The best pipeline is selected on the test set among $k \in \{1, 4\}$ below 2,000 labels. Bold marks the best value per row. The RoBERTa-large rows are being rerun with a lower learning rate.

Dataset	Embedding	Classifier	α	CICLe	few-shot	Δ	coverage (%)	set size	no LLM call (%)
Yahoo Answers	contriever	LR	0.05	57.8	58.1	-0.28 ^{n.s.} [-0.74, +0.20]	95.6	5.32	5.8
	minilm	LR	0.05	59.4	59.8	-0.37 ^{n.s.} [-0.82, +0.08]	95.0	5.67	3.4
	minilm	SVM	0.05	59.8	59.8	-0.05 ^{n.s.} [-0.50, +0.40]	95.6	5.79	4.7
	tfidf	LR	0.05	54.5	55.2	-0.69 [-1.10, -0.30]	96.1	7.97	1.1
	minilm	LR	0.01	59.8	59.8	-0.07 ^{n.s.} [-0.30, +0.17]	99.1	8.98	0.0
	minilm	LR	0.10	60.8	59.8	+0.93 [+0.31, +1.56]	90.1	4.01	9.8
	minilm	LR	0.20	63.4	59.8	+3.57 [+2.67, +4.48]	79.3	2.08	35.4
SST-5	contriever	LR	0.05	47.0	43.3	+3.69 [+2.82, +4.53]	95.5	3.34	0.5
	minilm	LR	0.05	45.0	42.3	+2.65 [+1.85, +3.46]	95.4	3.86	0.0
	minilm	SVM	0.05	44.9	42.3	+2.53 [+1.70, +3.34]	96.1	3.81	0.0
	tfidf	LR	0.05	42.8	41.2	+1.61 [+0.92, +2.30]	95.0	4.30	0.1
	minilm	LR	0.01	43.4	42.3	+1.05 [+0.64, +1.47]	99.2	4.67	0.0
	minilm	LR	0.10	45.1	42.3	+2.75 [+1.70, +3.76]	90.5	3.38	0.3
	minilm	LR	0.20	44.9	42.3	+2.52 [+1.24, +3.82]	80.3	2.71	3.0
SemEval-18	contriever	LR	0.05	16.3	15.7	+0.56 [+0.18, +0.95]	94.2	16.77	0.0
	minilm	LR	0.05	15.8	15.3	+0.47 [+0.13, +0.81]	94.4	17.12	0.0
	minilm	SVM	0.05	16.0	15.3	+0.69 [+0.34, +1.04]	94.9	17.13	0.0
	tfidf	LR	0.05	15.1	15.0	+0.11 ^{n.s.} [-0.15, +0.37]	95.1	18.06	0.0
	minilm	LR	0.01	15.4	15.3	+0.12 ^{n.s.} [-0.07, +0.30]	98.8	19.42	0.0
	minilm	LR	0.10	16.4	15.3	+1.04 [+0.56, +1.52]	88.5	14.64	0.0
	minilm	LR	0.20	16.7	15.3	+1.36 [+0.73, +1.96]	77.8	11.48	0.0
GoEmotions	contriever	LR	0.05	25.1	22.6	+2.46 [+1.67, +3.19]	95.0	21.07	0.0
	minilm	LR	0.05	23.3	22.4	+0.92 [+0.31, +1.56]	95.4	22.10	0.0
	minilm	SVM	0.05	23.7	22.4	+1.26 [+0.64, +2.00]	95.1	21.75	0.0
	tfidf	LR	0.05	23.1	22.5	+0.56 [+0.04, +1.13]	94.5	23.74	0.0
	minilm	LR	0.01	22.9	22.4	+0.49 [+0.14, +0.82]	98.8	26.74	0.0
	minilm	LR	0.10	23.8	22.4	+1.45 [+0.74, +2.33]	90.9	18.10	0.0
	minilm	LR	0.20	25.4	22.4	+3.02 [+2.09, +4.24]	81.0	13.15	0.0

Table 12: Ablation of the embedding, the classifier and α with Llama-3.1-8B and Llama-3.2-3B, both retrieval variants, $k \in \{1, 4\}$ and three seeds (24 cells). Few-shot prompting retrieves with the same embedding as the CICLe run it is paired with. Δ = CICLe minus few-shot with its 95% bootstrap interval. Coverage, set size and the share of instances answered without an LLM call describe the candidate sets.

Model	Dataset	seeds	Zero-shot	Fixed		Per-Class		Δ Fixed	Δ PC
				$k=1$	$k=4$	$k=1$	$k=4$		
Mistral-Nemo-12B	Yahoo Answers	42,43,44	57.8	56.8 / 57.2	61.0 / 61.3	57.6 / 59.6	63.4 / 63.0	+0.42 ^{n.s.} [-0.40, +1.24] (6)	+0.78 ^{n.s.} [-0.06, +1.62] (6)
Mistral-Nemo-12B	SST-5	42,43,44	27.5	37.6 / 42.5	41.6 / 45.1	41.7 / 45.6	42.0 / 46.9	+4.20 [+3.08, +5.26] (6)	+4.39 [+3.14, +5.66] (6)
Mistral-Nemo-12B	SemEval-18	42,43,44	16.6	17.8 / 18.2	19.7 / 20.3	19.4 / 19.9	20.9 / 20.8	+0.50 ^{n.s.} [-0.03, +1.02] (6)	+0.18 ^{n.s.} [-0.45, +0.83] (6)
Mistral-Nemo-12B	GoEmotions	42,43,44	27.2	25.9 / 27.0	28.4 / 28.4	27.1 / 28.3	30.3 / 30.7	+0.53 ^{n.s.} [-0.45, +1.48] (6)	+0.77 ^{n.s.} [-0.33, +2.05] (6)
Qwen2.5-32B	Yahoo Answers	42,43,44	63.2	64.0 / 65.0	66.4 / 66.9	63.9 / 65.2	65.8 / 66.4	+0.72 [+0.15, +1.29] (6)	+0.94 [+0.29, +1.60] (6)
Qwen2.5-32B	SST-5	42,43,44	49.3	52.1 / 53.8	53.5 / 54.8	55.5 / 55.9	55.2 / 55.5	+1.53 [+0.49, +2.58] (6)	+0.32 ^{n.s.} [-0.68, +1.29] (6)
Qwen2.5-32B	SemEval-18	42,43,44	15.4	17.3 / 17.6	19.2 / 19.8	19.8 / 20.2	20.2 / 20.4	+0.48 ^{n.s.} [-0.08, +1.03] (6)	+0.34 ^{n.s.} [-0.29, +0.93] (6)
Qwen2.5-32B	GoEmotions	42,43,44	31.5	31.2 / 31.4	32.7 / 32.4	30.7 / 30.5	32.4 / 32.2	-0.05 ^{n.s.} [-1.01, +0.93] (6)	-0.20 ^{n.s.} [-1.01, +0.64] (6)

Smallest models with CICLE against the largest model without examples (seed 42)
Yahoo Answers, best 3B + CICLE PC $k=4$, seed 42: Llama-3.2-3B 61.5, Qwen2.5-32B zero-shot, seed 42: 60.7
SST-5, best 3B + CICLE PC $k=4$, seed 42: Ministral-3B 51.0, Qwen2.5-32B zero-shot, seed 42: 48.8
SemEval-18, best 3B + CICLE PC $k=4$, seed 42: Ministral-3B 14.9, Qwen2.5-32B zero-shot, seed 42: 15.3
GoEmotions, best 3B + CICLE PC $k=4$, seed 42: Qwen2.5-3B 24.4, Qwen2.5-32B zero-shot, seed 42: 32.0

Table 13: Mistral-NeMo-12B and Qwen2.5-32B with and without CICLE. Cells give few-shot / CICLE macro-F1, mean over the seeds listed. Δ = CICLE minus few-shot over $k \in \{1, 4\}$, paired within the model, with 95% bootstrap intervals and the number of cells. The last rows compare the best 3B model with CICLE (Per-Class, $k = 4$) against the 32B model’s zero-shot score on seed 42.

Variant	Retr.	pairs	fixed	broken	both right (% of pairs)	both wrong	gold outside set			few-shot answer outside set		
							share	fixed	broken	share	fixed	broken
Yahoo Answers	Per-Class	18,000	4.7	3.4	58.0	33.8	5.0	0.0	4.1	5.0	39.5	4.1
Yahoo Answers	Fixed	18,000	4.2	4.3	53.7	37.8	5.0	0.0	5.6	7.8	31.1	3.5
SST-5	Per-Class	18,000	6.9	6.1	42.2	44.7	4.6	0.6	35.6	6.3	34.9	26.1
SST-5	Fixed	18,000	7.0	5.8	40.4	46.8	4.6	0.8	33.2	6.2	35.0	24.9
SemEval-18	Per-Class	18,000	3.1	3.2	18.8	74.9	5.6	0.0	11.1	7.6	10.1	8.2
SemEval-18	Fixed	18,000	2.7	2.6	18.5	76.2	5.6	0.0	8.8	10.3	7.0	4.8
GoEmotions	Per-Class	18,000	4.6	3.2	22.6	69.6	4.6	0.0	8.3	8.7	15.2	4.4
GoEmotions	Fixed	18,000	4.1	3.0	24.1	68.8	4.6	0.0	7.6	10.3	14.0	3.4
Ohsumed	Fixed	18,000	7.2	4.9	52.2	35.7	3.6	0.0	15.4	12.9	33.5	4.3
Yahoo Answers 100×	Per-Class	18,000	3.7	2.6	52.3	41.4	3.3	0.0	9.9	5.5	27.7	5.8
Yahoo Answers 100×	Fixed	18,000	3.5	3.0	49.6	44.0	3.3	0.0	9.5	7.0	23.1	4.5
SST-5 100×	Per-Class	18,000	4.7	4.1	39.7	51.6	4.0	0.0	40.8	5.1	34.4	32.2
SST-5 100×	Fixed	18,000	5.1	4.5	37.2	53.2	4.0	0.7	40.4	5.8	35.0	27.9
Yahoo Answers (renamed)	Per-Class	18,000	9.2	5.6	43.4	41.8	5.2	0.0	7.6	11.7	41.1	3.4
Yahoo Answers (renamed)	Fixed	18,000	8.8	5.2	36.0	50.0	5.2	0.0	7.0	18.4	30.5	2.0
SST-5 (renamed)	Per-Class	18,000	7.6	6.1	32.2	54.1	4.6	0.0	33.0	8.1	34.8	18.8
SST-5 (renamed)	Fixed	18,000	7.6	4.6	28.8	59.0	4.6	0.0	18.8	9.8	32.3	8.8
SemEval-18 (renamed)	Per-Class	18,000	3.7	2.9	10.7	82.7	5.6	0.0	7.6	6.2	13.3	6.8
SemEval-18 (renamed)	Fixed	18,000	2.8	2.0	9.3	85.9	5.6	0.0	5.9	13.4	7.7	2.4
GoEmotions (renamed)	Per-Class	18,000	5.4	3.7	14.4	76.5	4.5	0.0	6.5	14.7	13.0	2.0
GoEmotions (renamed)	Fixed	18,000	4.2	3.6	13.1	79.1	4.5	0.0	5.0	14.2	11.7	1.6
Ohsumed (renamed)	Fixed	18,000	8.7	6.3	40.7	44.3	3.7	0.0	11.6	20.6	29.7	2.1

Table 14: Fixes and breaks. CICLE at $k = 4$ against few-shot prompting at $k = 4$ with the same retrieval variant, six models, three seeds and 1,000 instances (18,000 pairs). Fixed = CICLE right and few-shot wrong, broken = the reverse. The right-hand blocks give the share of pairs whose gold label, or whose few-shot answer, lies outside the candidate set, and the fixed and broken rates within that subset.

Model	Yahoo Answers	SST-5	SemEval-18		GoEmotions	Ohsumed
			mean (max) invalid rate in %: zero-shot / few-shot / CICLE			
Llama-3.2-3B	1.7 (1.9) / 0.2 (0.7) / 0.3 (1.3)	0.0 (0.0) / 0.0 (0.1) / 0.0 (0.1)	0.4 (0.5) / 1.5 (5.1) / 1.5 (6.1)		0.2 (0.2) / 0.6 (1.3) / 0.4 (0.8)	2.5 (2.7) / 0.6 (1.3) / 0.1 (0.7)
Ministral-3B	1.6 (2.0) / 0.3 (2.0) / 0.2 (0.6)	0.0 (0.0) / 0.0 (0.0) / 0.0 (0.0)	0.1 (0.2) / 0.6 (1.3) / 0.4 (0.9)		0.1 (0.1) / 0.2 (0.5) / 0.1 (0.3)	0.6 (0.7) / 0.8 (1.4) / 0.3 (0.6)
Qwen2.5-3B	11.7 (13.2) / 9.9 (15.7) / 4.7 (10.2)	0.0 (0.0) / 0.0 (0.1) / 0.0 (0.1)	9.1 (9.9) / 14.6 (24.5) / 13.5 (22.4)		4.1 (4.9) / 3.9 (6.7) / 3.4 (5.9)	22.2 (23.7) / 20.7 (30.3) / 6.6 (8.9)
Mistral-7B	10.4 (10.8) / 2.7 (7.9) / 1.9 (7.1)	0.1 (0.1) / 0.0 (0.1) / 0.1 (0.8)	7.4 (7.9) / 5.2 (8.9) / 4.6 (7.8)		7.5 (8.0) / 5.2 (10.5) / 3.7 (6.9)	2.9 (3.4) / 3.1 (5.9) / 0.5 (0.9)
Qwen2.5-7B	11.1 (11.4) / 2.6 (5.0) / 1.7 (6.8)	0.0 (0.0) / 0.0 (0.2) / 0.0 (0.2)	12.3 (13.4) / 6.4 (9.9) / 6.6 (12.5)		5.5 (6.3) / 6.2 (7.8) / 5.0 (6.7)	2.5 (2.8) / 3.2 (6.5) / 0.9 (1.5)
Llama-3.1-8B	0.4 (0.5) / 0.1 (0.3) / 0.0 (0.1)	0.0 (0.0) / 0.0 (0.0) / 0.0 (0.0)	1.1 (1.4) / 0.8 (1.3) / 0.7 (1.2)		0.9 (1.0) / 0.8 (1.4) / 0.6 (1.1)	0.7 (1.0) / 0.2 (0.6) / 0.0 (0.1)
Mistral-Nemo-12B	4.1 (4.4) / 0.7 (2.2) / 0.2 (0.8)	0.0 (0.0) / 0.0 (0.0) / 0.0 (0.1)	5.6 (5.8) / 2.3 (5.5) / 2.4 (5.4)		1.0 (1.2) / 1.0 (2.2) / 0.8 (1.3)	— / — / —
Qwen2.5-32B	3.2 (3.7) / 0.5 (1.1) / 0.4 (1.1)	0.0 (0.0) / 0.0 (0.0) / 0.0 (0.0)	13.0 (14.2) / 6.0 (9.0) / 5.2 (7.9)		2.8 (3.1) / 2.7 (4.0) / 1.8 (2.7)	— / — / —

Table 15: Invalid (unparseable) outputs per model and dataset at the reference setting, mean and maximum over the (variant, k , seed) runs of each method.

Model	invalid outputs	most frequent raw outputs (count)
Llama-3.2-3B	1,392	discomfort (90), [emoji] (81), Neurological Diseases (77), [emoji] (69), I cannot predict the class for (40), [emoji] (39), [emoji] (35), [emoji] (30)
Ministral-3B	704	Religion (52), [emoji] *(Note: While not in your provided (41), Mycoses (40), **Mycoses** (18), [emoji] *(Note: The provided emoji list (11), [emoji] (10), [emoji] *(Note: The provided emoji list (9), nostalgia *(not listed in your (9)
Qwen2.5-3B	18,367	[emoji] (1,338), Neurological Diseases (1,328), [emoji] (1,023), Relationships (666), [emoji] (648), [emoji] (617), [emoji] (565), [emoji] (528)
Mistral-7B	7,322	Based on the provided text (106), It is difficult to accurately (84), appreciation (84), concern (81), [emoji] (77), discrimination (69), [emoji] (This emoji represents a party (61), anxiety (60)
Qwen2.5-7B	8,853	[emoji] (758), disbelief (418), advice (311), discomfort (265), Neurological Diseases (235), [emoji] (220), Religion (213), [emoji] (168)
Llama-3.1-8B	870	anxiety (87), discomfort (65), [emoji] (64), concern (40), nostalgia (30), Neurological Diseases (25), [emoji] (21), nostalgia is not in the list (19)
Mistral-Nemo-12B	1,207	[emoji] (94), gratuity (54), History (46), Religion (46), [emoji] (42), disbelief (37), [emoji] (36), [emoji] (36)
Qwen2.5-32B	2,561	[emoji] (319), [emoji] (103), disbelief (88), [emoji] (85), [emoji] (79), nostalgia It seems that the (65), disbelief However, since "dis (37), [emoji] (35)

Table 16: The most frequent raw outputs that matched no label (first 60 characters), pooled over the six main datasets and all methods at the reference setting. Emoji are shown as [emoji].

Case	Text	Gold	Candidate set	Few-shot	CICLe
fixed: few-shot answered outside the set	My favourite one is the wings on his back. I know his got 9 altogether, but what are they?	Sports	Education, Entertainment, Sports	Culture	Sports
fixed: few-shot answered outside the set	I have a relatively new Dell and it has some spots on it. I don't know what's safe to use....	Tech	Tech	Health	(no LLM call)
fixed: few-shot answered outside the set	nothing seems to work!	Tech	Business, Culture, Education, Entertainment, Health, Science, Tech	Family	Tech
broken: gold outside the set	I just started celebrating holidays my son is turning one soon. I have never attended a child's birthday party. Does anyone have ideas? He (...)	Family	Culture, Education, Entertainment, Health, Science, Sports, Tech	Family	Education
broken: gold outside the set	I Help with my home work ?Why do we argue about gay marriage? why people argue this? and give 3 reasons why we argue and explain?and conclus...	Family	Business, Culture, Education, Entertainment, Politics, Science, Sports, Tech	Family	Education
broken: gold outside the set	Can we ever compensate for the love and care they gave us in our childhood and how many sacrifices they did towards our upbringing?	Family	Culture, Education, Politics	Family	Culture
broken: switched inside the set	Hi ladies, Just curious when and HOW did you know if you were a lesbian; were you ashamed??	Family	Business, Culture, Education, Entertainment, Family, Health, Tech	Family	Culture
broken: switched inside the set	Who can tell the names of atleast seven characters? if you can you win...nothing you just get points for answering	Entertainment	Culture, Education, Entertainment, Science, Sports, Tech	Entertainment	Education
broken: switched inside the set	I watched it last night and it was the first episode...I hope and pray that the black woman who is preaching the "Words of God"...remind you...	Culture	Culture, Entertainment	Culture	Entertainment
fixed: few-shot answered outside the set	Help, help, I'm being repressed!	fear	annoyance, caring, confusion, desire, disappointment, disgust, embarrassment, fear, grief, nervousness, neutral, optimism, pride, realization, relief, remorse, sadness	anger	fear
fixed: few-shot answered outside the set	Lawson's Sip of Sunshine! Excellent Vermont beer.	admiration	admiration, amusement, caring, curiosity, desire, disappointment, disgust, embarrassment, excitement, fear, gratitude, grief, joy, love, nervousness, neutral, optimism, pride, realization, relief, remorse, sadness, surprise	approval	admiration
fixed: few-shot answered outside the set	Stuck on icy stairs.	neutral	admiration, annoyance, caring, confusion, curiosity, desire, disappointment, disgust, embarrassment, excitement, fear, grief, joy, nervousness, neutral, optimism, pride, realization, relief, remorse, surprise	disapproval	neutral
broken: gold outside the set	Please treat me with modern medicine. I have gastroparisis.	caring	admiration, annoyance, approval, curiosity, disappointment, disapproval, disgust, excitement, fear, grief, joy, love, nervousness, neutral, optimism, pride, realization, relief, remorse, sadness, surprise	caring	approval
broken: gold outside the set	Wasnt much to ruin.	neutral	admiration, amusement, anger, annoyance, approval, caring, confusion, curiosity, desire, disappointment, disapproval, disgust, embarrassment, excitement, fear, gratitude, grief, joy, nervousness, optimism, pride, realization, relief, remorse, surprise	neutral	amusement
broken: gold outside the set	Someone with erectile dysfunction already can't (conventionally) orgasm. What the fuck are you talking about?	anger	admiration, amusement, annoyance, approval, confusion, curiosity, desire, disappointment, disapproval, disgust, embarrassment, fear, grief, joy, love, nervousness, neutral, optimism, pride, realization, relief, remorse, sadness, surprise	anger	disapproval
broken: switched inside the set	Hope gallery for sure, had [NAME] do mine and a good friend back in 2015. Not sure if he's still there, however.	optimism	admiration, amusement, approval, caring, confusion, curiosity, desire, disappointment, disgust, embarrassment, excitement, fear, gratitude, grief, joy, love, nervousness, optimism, pride, realization, relief, remorse, surprise	optimism	curiosity
broken: switched inside the set	Tu eres very nice man	admiration	admiration, amusement, anger, annoyance, approval, confusion, disappointment, disgust, embarrassment, excitement, fear, gratitude, grief, joy, love, nervousness, optimism, pride, realization, relief, remorse, sadness, surprise	admiration	gratitude
broken: switched inside the set	I've exposed on social media. It made me feel better so I don't care what anyway has to say about it.	relief	admiration, amusement, anger, annoyance, approval, caring, curiosity, disappointment, disapproval, disgust, embarrassment, excitement, fear, gratitude, grief, nervousness, neutral, optimism, pride, realization, relief, remorse, sadness, surprise	relief	optimism

Table 17: Qualitative examples from Yahoo Answers and GoEmotions, Llama-3.1-8B, seed 42, Per-Class $k = 4$. Three instances per case, sampled with a fixed seed, texts truncated to 140 characters.

Dataset	Imbalance	Seed	class: training count (descending)
Yahoo Answers	10×	42	Sports 391, Business 303, Culture 234, Entertainment 182, Education 141, Health 109, Tech 84, Politics 66, Science 50, Family 39
Yahoo Answers	10×	43	Health 391, Family 303, Politics 234, Sports 182, Entertainment 141, Science 109, Tech 84, Business 66, Culture 50, Education 39
Yahoo Answers	10×	44	Education 391, Culture 303, Health 234, Sports 182, Tech 141, Science 109, Family 84, Business 66, Politics 50, Entertainment 39
Yahoo Answers	100×	42	Sports 645, Business 386, Culture 232, Entertainment 139, Education 83, Health 50, Tech 30, Politics 18, Science 10, Family 6
Yahoo Answers	100×	43	Health 645, Family 386, Politics 232, Sports 139, Entertainment 83, Science 50, Tech 30, Business 18, Culture 10, Education 6
Yahoo Answers	100×	44	Education 645, Culture 386, Health 232, Sports 139, Tech 83, Science 50, Family 30, Business 18, Politics 10, Entertainment 6
SST-5	10×	42	very positive 742, neutral 418, positive 234, negative 132, very negative 74
SST-5	10×	43	very positive 742, negative 418, neutral 234, very negative 132, positive 74
SST-5	10×	44	neutral 742, positive 418, very negative 234, very positive 132, negative 74
SST-5	100×	42	positive 1098, very negative 347, negative 110, neutral 34, very positive 11
SST-5	100×	43	neutral 1098, positive 347, very negative 110, negative 34, very positive 11
SST-5	100×	44	neutral 1098, positive 347, very negative 110, very positive 34, negative 11

Table 18: Class order and training-pool counts (after the 80/20 calibration split) under controlled imbalance for every seed. The class order is a permutation drawn from the seed.

Dataset	original name → nonsense word
Yahoo Answers	Business → zovobi, Culture → pomini, Education → repeke, Entertainment → varipu, Family → fidazi, Health → nudimi, Politics → reponu, Science → kavuba, Sports → totaro, Tech → zigita
GoEmotions	admiration → zovobi, amusement → pomini, anger → repeke, annoyance → varipu, approval → fidazi, caring → nudimi, confusion → reponu, curiosity → kavuba, desire → totaro, disappointment → zigita, disapproval → gugeve, disgust → vunadi, embarrassment → podipi, excitement → tapizu, fear → gupuko, gratitude → duvolu, grief → gifeze, joy → busina, love → defaza, nervousness → tusozu, neutral → kuveze, optimism → suzori, pride → nosidi, realization → ranusi, relief → zegati, remorse → delelo, sadness → zadeze, surprise → busura

Table 19: Nonsense-word mapping used for label renaming, the same for all seeds. SST-5, SemEval-18 and Ohsumed use the same word list in alphabetical order of the original names.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	65.4	66.0	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	63.2	63.6	0.0	0.0	0
Zero-shot	–	–	18	55.3	54.6	6.1	0.0	221
Few-shot	Fixed	1	18	54.8	54.0	4.8	1.0	285
Few-shot	Fixed	2	18	56.4	55.8	3.8	2.0	343
Few-shot	Fixed	4	18	58.4	58.0	2.8	4.0	455
Few-shot	Fixed	8	18	59.8	59.7	1.7	8.0	674
Few-shot	Per-Class	1	18	57.9	57.5	2.9	10.0	700
Few-shot	Per-Class	2	18	59.5	59.3	2.0	20.0	1,183
Few-shot	Per-Class	4	18	61.6	61.5	1.6	40.0	2,166
Few-shot	Per-Class	8	18	62.2	62.1	1.5	80.0	4,168
CICLe	Fixed	0	18	56.3	55.8	3.3	0.0	212
CICLe	Fixed	1	18	54.2	53.7	2.7	1.0	276
CICLe	Fixed	2	18	56.5	56.0	2.1	2.0	334
CICLe	Fixed	4	18	58.2	57.9	1.2	4.0	447
CICLe	Fixed	8	18	59.7	59.6	0.7	8.0	666
CICLe	Per-Class	1	18	59.5	59.2	1.3	5.8	503
CICLe	Per-Class	2	18	60.9	60.8	0.8	11.7	792
CICLe	Per-Class	4	18	62.7	62.7	0.6	23.3	1,378
CICLe	Per-Class	8	18	63.8	63.7	0.6	46.7	2,566
Top- m	Fixed	1	18	53.9	53.5	2.5	1.0	277
Top- m	Fixed	4	18	57.8	57.5	1.0	4.0	448
Top- m	Per-Class	1	18	58.5	58.2	1.3	5.7	505
Top- m	Per-Class	4	18	62.2	62.1	0.5	22.7	1,381
Prob. mass	Fixed	1	18	54.0	53.4	3.7	1.0	281
Prob. mass	Fixed	4	18	57.7	57.4	2.1	4.0	452
Prob. mass	Per-Class	1	18	58.0	57.7	2.1	7.8	604
Prob. mass	Per-Class	4	18	61.5	61.5	1.0	31.4	1,777
Marginal CP	Fixed	1	18	54.4	54.0	2.5	1.0	275
Marginal CP	Fixed	4	18	58.1	57.8	1.2	4.0	445
Marginal CP	Per-Class	1	18	59.1	59.0	1.2	6.0	509
Marginal CP	Per-Class	4	18	62.4	62.4	0.7	23.8	1,401
Oracle	Fixed	1	18	61.9	61.2	3.0	1.0	277
Oracle	Fixed	4	18	65.4	65.2	1.2	4.0	446
Oracle	Per-Class	1	18	66.5	66.2	1.4	5.7	492
Oracle	Per-Class	4	18	69.9	69.9	0.6	22.7	1,333

Table 20: Full grid for Yahoo Answers at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	35.5	40.8	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	45.5	49.4	0.0	0.0	0
Zero-shot	–	–	18	40.3	44.7	0.0	0.0	184
Few-shot	Fixed	1	18	40.6	42.8	0.0	1.0	217
Few-shot	Fixed	2	18	43.2	45.1	0.0	2.0	244
Few-shot	Fixed	4	18	44.1	46.2	0.0	4.0	298
Few-shot	Fixed	8	18	45.8	48.0	0.0	8.0	407
Few-shot	Per-Class	1	18	44.9	47.3	0.0	5.0	327
Few-shot	Per-Class	2	18	46.4	48.3	0.0	10.0	466
Few-shot	Per-Class	4	18	46.6	48.4	0.0	20.0	746
Few-shot	Per-Class	8	18	46.3	48.0	0.0	40.0	1,311
CICLe	Fixed	0	18	45.1	47.1	0.1	0.0	181
CICLe	Fixed	1	18	43.0	44.4	0.0	1.0	214
CICLe	Fixed	2	18	45.2	46.5	0.0	2.0	241
CICLe	Fixed	4	18	45.8	47.4	0.0	4.0	295
CICLe	Fixed	8	18	46.8	48.7	0.0	8.0	404
CICLe	Per-Class	1	18	46.2	48.0	0.0	3.9	292
CICLe	Per-Class	2	18	47.6	49.1	0.0	7.7	399
CICLe	Per-Class	4	18	47.5	49.1	0.0	15.5	615
CICLe	Per-Class	8	18	47.1	48.6	0.0	30.9	1,050
Top- m	Fixed	1	18	43.3	45.7	0.0	1.0	214
Top- m	Fixed	4	18	45.8	48.7	0.0	4.0	295
Top- m	Per-Class	1	18	46.0	48.7	0.0	4.0	296
Top- m	Per-Class	4	18	47.2	49.8	0.0	16.0	629
Prob. mass	Fixed	1	18	41.8	43.8	0.0	1.0	215
Prob. mass	Fixed	4	18	44.9	47.1	0.0	4.0	296
Prob. mass	Per-Class	1	18	45.2	47.5	0.0	4.5	312
Prob. mass	Per-Class	4	18	47.1	49.1	0.0	18.0	688
Marginal CP	Fixed	1	18	43.8	45.7	0.0	1.0	213
Marginal CP	Fixed	4	18	46.2	48.6	0.0	4.0	294
Marginal CP	Per-Class	1	18	46.4	48.9	0.0	3.7	286
Marginal CP	Per-Class	4	18	47.4	49.7	0.0	14.7	592
Oracle	Fixed	1	18	49.9	51.3	0.0	1.0	214
Oracle	Fixed	4	18	53.1	55.0	0.0	4.0	296
Oracle	Per-Class	1	18	53.6	55.6	0.0	4.0	297
Oracle	Per-Class	4	18	55.3	57.2	0.0	16.0	632

Table 21: Full grid for SST-5 at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	9.7	26.4	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	23.8	38.5	0.0	0.0	0
Zero-shot	–	–	18	12.2	19.7	5.1	0.0	236
Few-shot	Fixed	1	18	14.3	20.0	7.6	1.0	266
Few-shot	Fixed	2	18	14.6	20.1	7.0	2.0	290
Few-shot	Fixed	4	18	15.3	21.1	6.0	4.0	338
Few-shot	Fixed	8	18	15.4	22.2	4.3	8.0	436
Few-shot	Per-Class	1	18	15.2	20.7	4.4	20.0	731
Few-shot	Per-Class	2	18	15.1	21.2	3.4	40.0	1,223
Few-shot	Per-Class	4	18	15.4	21.9	2.9	80.0	2,209
Few-shot	Per-Class	8	18	15.7	23.1	3.2	160.0	4,189
CICLe	Fixed	0	18	13.0	19.6	5.0	0.0	225
CICLe	Fixed	1	18	14.7	19.9	7.2	1.0	255
CICLe	Fixed	2	18	15.1	20.1	6.7	2.0	279
CICLe	Fixed	4	18	15.9	21.2	5.5	4.0	328
CICLe	Fixed	8	18	16.1	22.3	4.1	8.0	425
CICLe	Per-Class	1	18	15.7	20.8	4.3	17.1	648
CICLe	Per-Class	2	18	15.6	21.3	3.1	34.2	1,069
CICLe	Per-Class	4	18	15.7	21.9	2.6	68.5	1,911
CICLe	Per-Class	8	18	16.0	22.9	2.8	137.0	3,603
Top- m	Fixed	1	18	14.9	20.3	6.9	1.0	254
Top- m	Fixed	4	18	15.7	21.2	5.1	4.0	327
Top- m	Per-Class	1	18	15.7	20.9	3.7	17.0	644
Top- m	Per-Class	4	18	15.7	22.1	2.3	68.0	1,897
Prob. mass	Fixed	1	18	14.9	20.3	6.9	1.0	254
Prob. mass	Fixed	4	18	15.6	21.2	5.0	4.0	326
Prob. mass	Per-Class	1	18	15.8	21.0	3.8	16.9	640
Prob. mass	Per-Class	4	18	15.6	22.1	2.3	67.5	1,883
Oracle	Fixed	1	18	15.8	21.6	7.4	1.0	255
Oracle	Fixed	4	18	17.0	23.1	5.9	4.0	328
Oracle	Per-Class	1	18	16.9	22.6	4.7	17.0	647
Oracle	Per-Class	4	18	17.2	24.0	2.8	68.0	1,904

Table 22: Full grid for SemEval-18 at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	11.2	43.7	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	32.2	50.9	0.0	0.0	0
Zero-shot	–	–	18	24.9	24.2	3.0	0.0	220
Few-shot	Fixed	1	18	24.6	24.6	3.8	1.0	246
Few-shot	Fixed	2	18	26.4	26.6	3.8	2.0	266
Few-shot	Fixed	4	18	26.4	27.2	3.3	4.0	305
Few-shot	Fixed	8	18	26.6	28.0	2.9	8.0	385
Few-shot	Per-Class	1	18	25.0	25.0	2.8	28.0	796
Few-shot	Per-Class	2	18	25.4	25.5	2.3	55.0	1,351
Few-shot	Per-Class	4	18	24.8	25.8	1.9	107.0	2,422
Few-shot	Per-Class	8	18	24.8	26.8	1.7	203.0	4,365
CICLe	Fixed	0	18	26.2	25.6	2.1	0.0	207
CICLe	Fixed	1	18	25.5	25.7	2.8	1.0	233
CICLe	Fixed	2	18	26.8	27.3	3.0	2.0	253
CICLe	Fixed	4	18	26.8	28.2	2.6	4.0	293
CICLe	Fixed	8	18	27.3	29.1	2.2	8.0	373
CICLe	Per-Class	1	18	26.0	26.4	2.2	22.1	665
CICLe	Per-Class	2	18	26.4	26.8	1.7	43.3	1,102
CICLe	Per-Class	4	18	26.0	27.2	1.6	83.7	1,942
CICLe	Per-Class	8	18	25.3	27.8	1.5	157.0	3,443
Top- m	Fixed	1	18	22.7	24.5	2.9	1.0	232
Top- m	Fixed	4	18	24.6	27.0	2.8	4.0	292
Top- m	Per-Class	1	18	23.9	25.7	2.4	22.0	664
Top- m	Per-Class	4	18	23.0	26.0	1.9	88.0	2,007
Prob. mass	Fixed	1	18	22.9	25.0	2.5	1.0	226
Prob. mass	Fixed	4	18	24.9	27.6	2.5	4.0	286
Prob. mass	Per-Class	1	18	24.3	26.2	2.0	18.8	592
Prob. mass	Per-Class	4	18	23.8	27.0	1.6	75.2	1,740
Marginal CP	Fixed	1	18	23.5	26.2	1.7	1.0	217
Marginal CP	Fixed	4	18	24.9	28.7	1.8	4.0	276
Marginal CP	Per-Class	1	18	24.8	27.7	1.7	14.4	490
Marginal CP	Per-Class	4	18	24.6	28.6	1.3	57.4	1,365
Oracle	Fixed	1	18	28.0	28.1	3.1	1.0	233
Oracle	Fixed	4	18	29.2	30.3	2.9	4.0	293
Oracle	Per-Class	1	18	28.3	28.6	2.4	22.0	661
Oracle	Per-Class	4	18	28.4	29.5	1.7	84.1	1,938

Table 23: Full grid for GoEmotions at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	45.2	65.0	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	60.8	72.4	0.0	0.0	0
Zero-shot	–	–	18	35.9	43.2	5.2	0.0	579
Few-shot	Fixed	1	18	46.7	52.7	5.3	1.0	897
Few-shot	Fixed	2	18	48.6	55.2	5.0	2.0	1,207
Few-shot	Fixed	4	18	50.4	57.1	3.9	4.0	1,826
Few-shot	Fixed	8	18	52.1	58.5	3.3	8.0	3,059
Few-shot	Per-Class	1	18	44.6	50.5	6.4	23.0	7,518
CICLe	Fixed	0	18	40.2	48.6	1.3	0.0	509
CICLe	Fixed	1	18	47.6	55.2	1.8	1.0	828
CICLe	Fixed	2	18	49.7	57.6	1.6	2.0	1,138
CICLe	Fixed	4	18	51.7	59.4	1.3	4.0	1,756
CICLe	Fixed	8	18	53.1	60.5	0.9	8.0	2,985
CICLe	Per-Class	1	18	49.8	57.3	1.5	12.3	4,159
Top- m	Fixed	1	18	46.9	53.0	2.7	1.0	831
Top- m	Fixed	4	18	50.3	57.1	2.0	4.0	1,760
Prob. mass	Fixed	1	18	46.2	51.9	3.6	1.0	843
Prob. mass	Fixed	4	18	50.1	56.3	2.7	4.0	1,771
Marginal CP	Fixed	1	18	45.5	50.7	0.9	1.0	782
Marginal CP	Fixed	4	18	48.1	53.6	0.7	4.0	1,704
Oracle	Fixed	1	18	55.9	62.0	2.6	1.0	826
Oracle	Fixed	4	18	60.3	66.9	1.8	4.0	1,754

Table 24: Full grid for Ohsumed at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	50.4	53.6	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	60.1	60.7	0.0	0.0	0
Few-shot	Fixed	1	18	54.2	53.4	4.8	1.0	285
Few-shot	Fixed	4	18	56.4	56.0	2.9	4.0	452
Few-shot	Per-Class	1	18	56.8	56.3	3.0	10.0	710
Few-shot	Per-Class	4	18	59.4	59.2	1.7	40.0	2,198
CICLe	Fixed	1	18	53.6	53.1	3.0	1.0	277
CICLe	Fixed	4	18	56.0	55.8	1.7	4.0	447
CICLe	Per-Class	1	18	58.2	57.9	1.7	6.2	539
CICLe	Per-Class	4	18	60.3	60.3	0.9	25.0	1,509
Top- m	Fixed	1	18	53.2	52.6	2.7	1.0	276
Top- m	Fixed	4	18	55.3	55.2	1.3	4.0	445
Top- m	Per-Class	1	18	56.1	55.8	1.3	6.0	518
Top- m	Per-Class	4	18	58.8	58.8	0.6	24.0	1,420
Prob. mass	Fixed	1	18	53.5	52.9	3.6	1.0	280
Prob. mass	Fixed	4	18	55.6	55.3	2.0	4.0	448
Prob. mass	Per-Class	1	18	56.4	56.0	1.9	7.5	591
Prob. mass	Per-Class	4	18	59.3	59.1	1.1	30.0	1,714
Marginal CP	Fixed	1	18	54.0	53.6	2.4	1.0	274
Marginal CP	Fixed	4	18	56.3	56.2	1.3	4.0	442
Marginal CP	Per-Class	1	18	57.6	57.5	1.2	5.3	485
Marginal CP	Per-Class	4	18	59.7	59.7	0.6	21.4	1,289
Oracle	Fixed	1	18	60.4	59.7	3.2	1.0	277
Oracle	Fixed	4	18	63.0	62.6	1.5	4.0	444
Oracle	Per-Class	1	18	64.9	64.7	1.4	6.0	515
Oracle	Per-Class	4	18	67.1	67.2	0.7	24.0	1,422

Table 25: Full grid for Yahoo Answers, $10\times$ imbalance at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	31.4	39.0	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	43.1	47.6	0.0	0.0	0
Few-shot	Fixed	1	18	52.8	52.2	4.8	1.0	284
Few-shot	Fixed	4	18	52.5	52.6	3.1	4.0	449
Few-shot	Per-Class	1	18	55.1	54.6	3.1	10.0	748
Few-shot	Per-Class	4	18	54.6	54.9	1.8	40.0	2,370
CICLe	Fixed	1	18	52.9	52.3	3.8	1.0	279
CICLe	Fixed	4	18	52.9	53.0	2.3	4.0	447
CICLe	Per-Class	1	18	56.0	55.9	2.2	7.7	635
CICLe	Per-Class	4	18	55.4	56.0	1.4	30.9	1,918
Top- m	Fixed	1	18	46.3	47.9	4.2	1.0	278
Top- m	Fixed	4	18	46.5	48.8	2.3	4.0	443
Top- m	Per-Class	1	18	48.1	50.0	2.5	7.7	597
Top- m	Per-Class	4	18	48.5	51.0	1.2	30.7	1,741
Prob. mass	Fixed	1	18	43.6	46.1	3.3	1.0	275
Prob. mass	Fixed	4	18	44.1	47.2	1.8	4.0	439
Prob. mass	Per-Class	1	18	46.0	49.0	1.9	6.0	511
Prob. mass	Per-Class	4	18	46.6	49.9	0.9	23.9	1,398
Marginal CP	Fixed	1	18	41.7	45.4	2.0	1.0	270
Marginal CP	Fixed	4	18	42.5	46.8	1.0	4.0	433
Marginal CP	Per-Class	1	18	43.7	48.0	1.0	4.1	420
Marginal CP	Per-Class	4	18	43.8	48.4	0.5	16.5	1,034
Oracle	Fixed	1	18	56.1	55.4	3.7	1.0	279
Oracle	Fixed	4	18	55.9	55.8	2.1	4.0	442
Oracle	Per-Class	1	18	59.1	58.7	2.2	7.7	622
Oracle	Per-Class	4	18	59.0	59.3	1.2	30.7	1,866

Table 26: Full grid for Yahoo Answers, $100\times$ imbalance at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	20.5	30.0	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	40.1	44.0	0.0	0.0	0
Few-shot	Fixed	1	18	40.0	42.4	0.0	1.0	217
Few-shot	Fixed	4	18	42.4	44.8	0.0	4.0	297
Few-shot	Per-Class	1	18	44.6	47.0	0.0	5.0	326
Few-shot	Per-Class	4	18	44.6	46.6	0.0	20.0	740
CICLe	Fixed	1	18	42.7	44.6	0.0	1.0	214
CICLe	Fixed	4	18	44.3	46.5	0.0	4.0	294
CICLe	Per-Class	1	18	46.3	48.5	0.0	4.1	298
CICLe	Per-Class	4	18	46.5	48.3	0.0	16.2	633
Top- m	Fixed	1	18	41.1	43.3	0.0	1.0	214
Top- m	Fixed	4	18	42.2	45.1	0.0	4.0	294
Top- m	Per-Class	1	18	43.9	46.5	0.0	4.0	296
Top- m	Per-Class	4	18	44.1	46.7	0.0	16.0	625
Prob. mass	Fixed	1	18	41.2	42.9	0.0	1.0	215
Prob. mass	Fixed	4	18	42.5	45.0	0.0	4.0	294
Prob. mass	Per-Class	1	18	44.7	46.8	0.0	4.1	300
Prob. mass	Per-Class	4	18	44.4	46.5	0.0	16.6	641
Marginal CP	Fixed	1	18	40.0	42.2	0.0	1.0	214
Marginal CP	Fixed	4	18	40.7	43.7	0.0	4.0	294
Marginal CP	Per-Class	1	18	42.4	45.0	0.0	3.6	286
Marginal CP	Per-Class	4	18	42.4	45.1	0.0	14.6	587
Oracle	Fixed	1	18	49.0	50.5	0.0	1.0	214
Oracle	Fixed	4	18	51.3	53.2	0.0	4.0	295
Oracle	Per-Class	1	18	53.3	55.2	0.0	4.0	297
Oracle	Per-Class	4	18	54.0	55.7	0.0	16.0	628

Table 27: Full grid for SST-5, $10\times$ imbalance at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
MiniLM + LR	–	–	3	12.6	23.0	0.0	0.0	0
RoBERTa-base (fine-tuned)	–	–	6	32.1	39.2	0.0	0.0	0
Few-shot	Fixed	1	18	37.7	40.9	0.0	1.0	217
Few-shot	Fixed	4	18	38.2	41.7	0.0	4.0	298
Few-shot	Per-Class	1	18	42.1	44.9	0.0	5.0	329
Few-shot	Per-Class	4	18	40.8	43.8	0.0	20.0	749
CICLe	Fixed	1	18	39.3	41.6	0.0	1.0	216
CICLe	Fixed	4	18	39.7	42.3	0.0	4.0	297
CICLe	Per-Class	1	18	43.4	45.7	0.0	4.4	309
CICLe	Per-Class	4	18	41.9	44.3	0.0	17.4	674
Top- m	Fixed	1	18	36.0	41.6	0.0	1.0	215
Top- m	Fixed	4	18	36.3	41.9	0.0	4.0	297
Top- m	Per-Class	1	18	38.9	44.4	0.0	4.3	311
Top- m	Per-Class	4	18	38.0	43.5	0.0	17.3	681
Prob. mass	Fixed	1	18	30.9	38.3	0.0	1.0	212
Prob. mass	Fixed	4	18	30.9	38.2	0.0	4.0	293
Prob. mass	Per-Class	1	18	32.1	39.5	0.0	2.9	265
Prob. mass	Per-Class	4	18	31.7	39.4	0.0	11.6	509
Marginal CP	Fixed	1	18	29.8	38.1	0.2	1.0	211
Marginal CP	Fixed	4	18	28.6	37.1	0.1	4.0	293
Marginal CP	Per-Class	1	18	30.1	38.5	0.0	2.4	251
Marginal CP	Per-Class	4	18	29.3	38.1	0.0	9.8	455
Oracle	Fixed	1	18	44.6	46.7	0.0	1.0	216
Oracle	Fixed	4	18	45.1	47.3	0.0	4.0	298
Oracle	Per-Class	1	18	48.7	50.8	0.0	4.3	309
Oracle	Per-Class	4	18	47.9	49.9	0.0	17.3	674

Table 28: Full grid for SST-5, 100 $\times$ imbalance at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
Few-shot	Fixed	1	18	25.3	26.1	0.6	1.0	300
Few-shot	Fixed	4	18	40.6	41.2	0.4	4.0	474
Few-shot	Per-Class	1	18	34.1	35.5	0.2	10.0	727
Few-shot	Per-Class	4	18	47.8	49.0	0.0	40.0	2,236
CICLe	Fixed	0	18	18.9	21.3	1.5	0.0	220
CICLe	Fixed	1	18	34.3	34.5	0.6	1.0	286
CICLe	Fixed	4	18	44.6	44.8	0.3	4.0	461
CICLe	Per-Class	1	18	41.1	41.3	0.2	5.8	521
CICLe	Per-Class	4	18	52.0	52.6	0.0	23.4	1,420
Top- m	Fixed	1	18	31.1	31.9	0.7	1.0	286
Top- m	Fixed	4	18	42.7	43.0	0.3	4.0	462
Top- m	Per-Class	1	18	37.4	37.9	0.3	5.7	521
Top- m	Per-Class	4	18	50.5	51.0	0.0	22.7	1,420
Prob. mass	Fixed	1	18	27.9	28.7	0.6	1.0	293
Prob. mass	Fixed	4	18	40.9	41.3	0.4	4.0	468
Prob. mass	Per-Class	1	18	35.7	36.5	0.3	7.8	625
Prob. mass	Per-Class	4	18	48.5	49.4	0.0	31.4	1,832
Marginal CP	Fixed	1	3	47.0	47.3	0.0	1.0	204
Marginal CP	Fixed	4	3	55.6	56.6	0.0	4.0	374
Marginal CP	Per-Class	1	3	43.8	44.8	0.0	6.0	439
Marginal CP	Per-Class	4	3	55.7	56.7	0.0	23.8	1,335
Oracle	Fixed	1	3	52.5	52.6	0.0	1.0	204
Oracle	Fixed	4	3	63.4	64.2	0.0	4.0	374
Oracle	Per-Class	1	3	47.3	48.4	0.0	5.7	421
Oracle	Per-Class	4	2	64.2	64.9	0.0	22.0	1,222

Table 29: Full grid for Yahoo Answers, renamed labels at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
Few-shot	Fixed	1	18	8.0	10.8	0.7	1.0	286
Few-shot	Fixed	4	18	11.8	16.7	1.3	4.0	351
Few-shot	Per-Class	1	18	10.6	12.4	0.5	28.0	874
Few-shot	Per-Class	4	18	14.4	18.1	0.5	107.0	2,609
CICLe	Fixed	0	18	2.1	3.5	1.1	0.0	238
CICLe	Fixed	1	18	8.8	11.6	1.0	1.0	266
CICLe	Fixed	4	18	12.3	17.3	1.1	4.0	330
CICLe	Per-Class	1	18	11.7	14.0	0.6	22.1	727
CICLe	Per-Class	4	18	15.3	19.8	0.6	83.8	2,091
Top- m	Fixed	1	18	8.1	11.5	2.5	1.0	265
Top- m	Fixed	4	18	12.1	17.7	1.4	4.0	329
Top- m	Per-Class	1	18	11.1	13.8	0.6	22.0	725
Top- m	Per-Class	4	18	14.5	19.1	0.6	88.0	2,161
Prob. mass	Fixed	1	18	8.1	11.8	3.1	1.0	254
Prob. mass	Fixed	4	18	12.1	17.6	1.5	4.0	318
Prob. mass	Per-Class	1	18	11.4	14.2	0.6	18.8	644
Prob. mass	Per-Class	4	18	14.8	20.0	0.6	75.2	1,872
Marginal CP	Fixed	1	2	10.9	15.0	0.2	1.0	159
Marginal CP	Fixed	4	2	16.9	28.9	0.1	4.0	221
Marginal CP	Per-Class	1	2	15.1	19.5	0.4	14.3	443
Marginal CP	Per-Class	4	1	18.1	27.8	0.3	56.2	1,313
Oracle	Fixed	1	1	10.7	13.6	0.0	1.0	185
Oracle	Fixed	4	1	18.1	28.9	0.1	4.0	247
Oracle	Per-Class	1	1	15.5	18.5	0.0	22.0	621
Oracle	Per-Class	4	1	21.0	28.3	1.1	84.0	1,943

Table 30: Full grid for GoEmotions, renamed labels at the reference setting. Means over the runs of the six small models (supervised rows over three seeds). A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
Few-shot	Fixed	1	18	24.3	26.5	0.6	1.0	223
Few-shot	Fixed	4	18	32.1	33.4	0.3	4.0	307
Few-shot	Per-Class	1	18	33.3	34.8	0.1	5.0	337
Few-shot	Per-Class	4	18	36.8	38.3	0.0	20.0	771
CICLe	Fixed	0	15	21.9	24.3	3.1	0.0	201
CICLe	Fixed	1	18	28.4	29.4	0.3	1.0	219
CICLe	Fixed	4	18	35.5	36.4	0.2	4.0	303
CICLe	Per-Class	1	18	35.4	36.4	0.1	3.9	300
CICLe	Per-Class	4	18	38.8	39.8	0.0	15.5	635
Top- m	Fixed	1	18	25.4	27.2	0.1	1.0	219
Top- m	Fixed	4	18	34.6	35.6	0.1	4.0	304
Top- m	Per-Class	1	18	35.4	36.4	0.1	4.0	305
Top- m	Per-Class	4	18	39.2	40.3	0.0	16.0	651
Prob. mass	Fixed	1	18	25.2	27.0	0.5	1.0	221
Prob. mass	Fixed	4	18	33.5	34.5	0.2	4.0	305
Prob. mass	Per-Class	1	18	34.4	35.5	0.1	4.5	321
Prob. mass	Per-Class	4	18	37.9	39.1	0.0	18.0	712
Marginal CP	Fixed	1	3	32.8	35.5	0.0	1.0	138
Marginal CP	Fixed	4	3	37.6	40.3	0.0	4.0	221
Marginal CP	Per-Class	1	3	35.4	38.7	0.0	3.7	213
Marginal CP	Per-Class	4	3	41.8	44.0	0.0	14.7	522
Oracle	Fixed	1	3	37.2	39.6	0.0	1.0	140
Oracle	Fixed	4	3	44.2	46.7	0.0	4.0	222
Oracle	Per-Class	1	3	42.8	45.5	0.0	4.0	223
Oracle	Per-Class	4	3	47.2	49.3	0.0	16.0	561

Table 31: Full grid for SST-5, renamed labels at the reference setting. Means over the runs of the six small models. A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
Few-shot	Fixed	1	18	6.3	8.4	6.0	1.0	267
Few-shot	Fixed	4	18	8.3	11.3	5.3	4.0	340
Few-shot	Per-Class	1	18	7.7	9.7	2.1	20.0	732
Few-shot	Per-Class	4	18	10.5	13.6	0.1	80.0	2,212
CICLe	Fixed	0	15	2.6	4.7	6.7	0.0	242
CICLe	Fixed	1	18	7.0	9.2	4.5	1.0	257
CICLe	Fixed	4	18	9.2	12.1	4.3	4.0	330
CICLe	Per-Class	1	18	8.3	10.4	1.9	17.2	651
CICLe	Per-Class	4	18	11.3	14.4	0.1	68.6	1,921
Top- m	Fixed	1	18	6.9	9.1	4.7	1.0	256
Top- m	Fixed	4	18	9.0	12.0	4.2	4.0	329
Top- m	Per-Class	1	18	8.4	10.7	2.1	17.0	647
Top- m	Per-Class	4	18	11.0	14.2	0.2	68.0	1,905
Prob. mass	Fixed	1	18	6.9	9.0	4.7	1.0	255
Prob. mass	Fixed	4	18	8.9	12.0	4.2	4.0	329
Prob. mass	Per-Class	1	18	8.4	10.6	2.0	16.9	643
Prob. mass	Per-Class	4	18	11.1	14.3	0.2	67.5	1,891
Marginal CP	Fixed	1	3	8.6	13.1	0.0	1.0	171
Marginal CP	Fixed	4	3	11.8	18.7	0.0	4.0	242
Marginal CP	Per-Class	1	3	11.4	16.9	0.0	16.0	524
Marginal CP	Per-Class	4	2	13.0	20.1	0.0	64.2	1,663
Oracle	Fixed	1	2	8.9	14.2	0.0	1.0	175
Oracle	Fixed	4	2	13.6	21.9	0.1	4.0	245
Oracle	Per-Class	1	2	12.0	18.2	0.0	17.0	550
Oracle	Per-Class	4	2	15.7	23.6	0.0	68.0	1,757

Table 32: Full grid for SemEval-18, renamed labels at the reference setting. Means over the runs of the six small models. A row with fewer than 18 runs is an incomplete configuration.

Method	Retrieval	k	runs	macro-F1	acc.	invalid (%)	shots	tokens
Few-shot	Fixed	1	18	27.1	32.0	2.4	1.0	829
Few-shot	Fixed	4	18	40.2	47.0	1.5	4.0	1,749
CICLe	Fixed	0	18	4.5	7.5	2.6	0.0	476
CICLe	Fixed	1	18	27.8	32.9	0.9	1.0	792
CICLe	Fixed	4	18	41.5	49.4	0.9	4.0	1,712
Top- m	Fixed	1	18	28.2	33.3	1.5	1.0	789
Top- m	Fixed	4	18	41.1	47.6	0.8	4.0	1,709
Prob. mass	Fixed	1	18	27.3	32.9	1.9	1.0	795
Prob. mass	Fixed	4	18	39.9	46.6	1.1	4.0	1,714
Marginal CP	Fixed	1	18	33.9	40.5	0.4	1.0	762
Marginal CP	Fixed	4	18	42.1	48.3	0.4	4.0	1,676
Oracle	Fixed	1	18	30.3	36.2	1.0	1.0	791
Oracle	Fixed	4	18	47.4	54.7	0.7	4.0	1,709

Table 33: Full grid for Ohsumed, renamed labels at the reference setting. Means over the runs of the six small models. A row with fewer than 18 runs is an incomplete configuration.

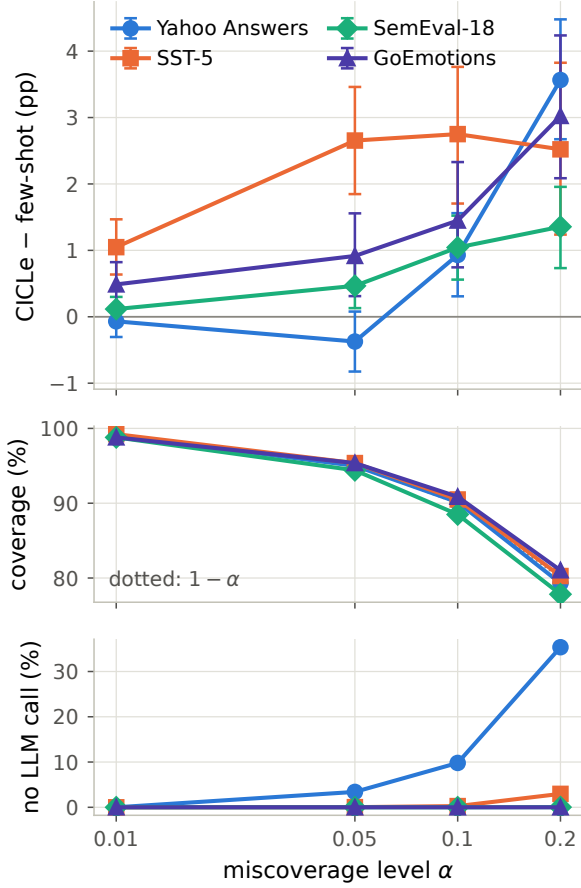


Figure 6: Effect of the miscoverage level α with Llama-3.1-8B and Llama-3.2-3B, both retrieval variants, $k \in \{1, 4\}$ and three seeds (24 cells per point). Top: CICLe minus few-shot with 95% bootstrap intervals. Middle: empirical coverage of the candidate sets against the $1 - \alpha$ target. Bottom: share of test instances answered by the base classifier without an LLM call.